Supplementary Online Content

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This supplementary material has been provided by the authors to give readers additional information about their work.

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eMethods

Data sources and searches

We searched MEDLINE, Embase, and the Cochrane Library from inception through 26 December 2021. The full search strategy is presented in online eTable S2-4. Briefly, we searched using variations of the exposure terms (low and non-nutritive sweetened beverages (LNCSBs), sugar-sweetened beverages (SSBs)), outcome terms (adiposity, glycemia, blood lipids, blood pressure, non-alcoholic fatty liver disease (NAFLD) and uric acid) and study design terms (randomized controlled trial, randomized, placebo). The search was limited to human studies with no language restriction. Additionally, reference lists of selected studies and reviews were searched, experts in the field were contacted and Google Scholar searches were conducted to identify any additional articles.

Study selection

Online eTables S5 shows the summary of our study selection using the PICOTS framework [1]. We included RCTs of at least 2-weeks duration that investigated the effect of one of three beverages (LNCSBs, SSBs or water) on cardiometabolic risk factors compared to another among the three beverages. Trials were included if the intervention arm assessed the effect of LNCSBs, SSBs or water consumed alone or while on a weight-loss or nutrition education program in adults. Trials were excluded if they had non-usual intake method (e.g., used low and non-nutritive sweeteners (LNCS) capsules that bypass the oral taste receptors); had a mixed intervention; had a duration of <2 weeks; did not use a comparator arm of LNCSBs, SSBs or water; included children, pregnant or breastfeeding women; or did not provide suitable end-point data. We only studied LNCS in liquid (beverage trials) as these allow for clear comparisons. Studies of LNCSs added to foods, fortified beverages or nutrient-dense beverages (e.g., milk, juice, etc.) were excluded due to the presence of other macronutrients. When multiple publications existed for the same study, the article with the most updated information was included. Published abstracts were not included.

Data extraction

Two independent reviewers (NM and RZ) assessed the titles and abstracts of all identified studies and reviewed and extracted relevant data from each report, including study design, blinding, sample size, participant characteristics, follow-up duration, nature of the interventions/comparators, doses, outcome data and funding source. In those trials where the data were only reported in figures, we extracted data using Plot Digitizer V.2.6.8 (http://plotdigitizer.sourceforge.net/). Additional information was requested from the authors when necessary. Disagreement were resolved by consensus or, when necessary, arbitration by TAK and JLS.

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Risk of bias assessment

Two independent reviewers (NM and RZ) assessed Risk of bias for each included trial using the Cochrane Risk of Bias tool [2]. Assessment was done across 5 domains of bias (sequence generation, allocation concealment, blinding of participants and personnel, incomplete outcome data and selective reporting). The risk of bias was assessed as either low (proper methods taken to reduce bias), high (improper methods creating bias) or unclear (insufficient information provided to determine the bias level).

Data synthesis

Mean differences between the arms and their respective variance terms were extracted and used as the basis for analysis for each trial comparison. If mean differences were not provided, they were derived from available data using published formulas.[3] When median data were reported, they were converted to mean data with corresponding variances using established methods.[4,5] When no variance data were available, the standard deviation was taken from a trial similar in size, participants, and nature of intervention. We used change from baseline values from each study to calculate the mean differences between treatments for each substitution (LNCSBs for SSBs, water for SSBs and LNCSBs for water), otherwise we used post-intervention values (Supplementary Data S1-S20). For crossover trials and for within arm changes in parallel trials, we used a correlation coefficient of 0.5 in pairwise analysis to calculate standard errors.[6-8] To mitigate a unit of analysis error, when arms of trials with multiple interventions or control arms were used more than once, the corresponding sample size was divided accordingly.[3] Non-HDL-C values that were not reported were derived by subtracting HDL-C from total cholesterol values with standard errors derived from HDL-C and total cholesterol variance data using the inverse variance law.[9] For trials in which the change in BMI was not reported, but body weight was reported, then if baseline BMI was available, these data were used to calculate the height, which could then be used to calculate the end BMI and change in BMI. The change in BMI variance was imputed using published formula[3] and a correlation coefficient of 0.5.[6-8]

Network meta-analysis, based on a frequentist framework, was conducted using the "network" suite of commands available in STATA version 15 (College Station, TX: StataCorp LP). We performed a random-effects network meta-analysis for each outcome to compare the three interventions simultaneously (LNCSBs, SSBs and water) in a single analysis by combining both direct and indirect evidence across the selected network of studies. The network meta-analysis synthesized all of the available evidence (direct and indirect effects) and quantified the pooled network effect of each intervention against every other intervention. We reported our results as mean differences (MDs) and 95% confidence intervals (CIs). To display the results for outcomes on the same plot, standardized mean differences (SMDs) and pseudo

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95% CIs were calculated, whereby the SMD 95% CIs were proportionally scaled to the MD 95% CIs. The network diagrams were generated to show the interactions among the studies included in the network meta-analysis and to illustrate the available direct comparisons between treatments [10].

Inconsistency was assessed in the direct, indirect, and network estimates. We assessed interstudy heterogeneity in the direct (each pairwise comparison arm) estimates using the Cochran Q statistic with quantification by the I^2 statistic, where an $I^2 \geq 50\%$, $P<0.10$ was considered an indication of substantial interstudy heterogeneity. We measured incoherence in the network estimates using both local (loop-specific and side-splitting) and global (design-by-treatment interaction model) approaches to evaluate the presence of incoherence. The loop-specific approach looked at the inconsistency in each closed loop in the network [10] while the side-splitting approach detected direct estimate comparisons that disagreed with the indirect evidence from the entire network [11]. The design-by-treatment interaction model was applied as a global approach to check simultaneously for inconsistency from all possible sources in the network [12]. If ≥10 comparisons were available for all comparisons, then we conducted a priori subgroup analyses by age, study duration, type of design, disease status, risk of bias and funding source.

Indirectness was assessed in the indirect comparisons by evaluation of intransitivity across the pairwise comparisons comprising the indirect estimates for important study characteristics including age, study length, sample size and percentage of males. Intransitivity was considered present if there was no overlap in the range between the pairwise comparisons.

Publication bias was assessed if ≥10 trial comparisons were available for all comparisons, We used comparison-adjusted funnel plots to assess funnel plot asymmetry[10]. Asymmetry around the line of the meta-analysis summary effect suggested evidence of small-study effects.

Grading of the evidence

We assessed the certainty of the evidence using the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) system [13] with an extension for network meta-analyses [14] and other recent guidance [15-17] from the GRADE Working group. Evidence was graded as high, moderate, low or very low certainty. Network estimates of RCTs and the direct and indirect estimates that make-up these network estimates started at high certainty of evidence and were downgraded by established criteria. These criteria included serious risk of bias (weight of RCTs in the direct estimates show high risk of bias assessed by the Cochrane Risk of Bias tool), inconsistency (substantial unexplained heterogeneity [$I^2 \geq 50\%$, $P_Q<0.10$] in the direct estimates and incoherence in the network estimates by the loop-specific [$p<0.05$], side-splitting [$p<0.05$] and design-by-treatment interaction model, $p<0.05$ or if by visual inspection the direct

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and indirect results were in opposite directions), indirectness (presence of factors related to the participants, interventions, and study conditions that limited the generalizability of the results among the direct estimates and intransitivity in the indirect estimates or if information for the network estimate was based upon only 1 direct study or only from indirect studies), imprecision (the 95% CIs for effect estimates were wide and crossed prespecified minimally important differences (MIDs) for benefit and/or harm in the direct and network estimates) and publication bias (evidence of comparison adjusted funnel plot asymmetry).

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eTables

eTable 1. Search strategy MEDLINE

Database	MEDLINE[1]		
Search Terms	1. aspartame.mp. 2. exp Aspartame/ 3. neotame.mp. 4. saccharin.mp. 5. exp Saccharin/ 6. sucralose.mp. 7. stevia.mp. 8. exp Stevia/ 9. acesulfame.mp. 10. exp Sweetening Agents/ 11. sugar substitute*.mp. 12. noncaloric.mp. 13. non-caloric.mp. 14. nonnutritive.mp. 15. non-nutritive.mp. 16. no calorie*.mp. 17. low calorie sweeten*.mp. 18. sugar-free.mp. 19. artificial sweet*.mp. 20. diet beverage*.mp. 21. diet soda.mp. 22. exp Carbonated Beverages/ 23. ssb.mp. 24. sugar*sweetened beverage*.mp. 25. Fructose/ 26. fructose.mp. 27. dietary sucrose/ 28. sucrose.mp. 29. High Fructose Corn Syrup/ 30. High Fructose Corn Syrup.mp. 31. Soft drink*.mp. 32. cola.mp. 33. sugar*.mp. 34. or/1-33 35. Blood Pressure/ 36. systolic blood pressure.mp. 37. SBP.mp. 38. diastolic blood pressure.mp. 39. DBP.mp. 40. 35 or 36 or 37 or 38 or 39 41. glyc*m*.mp. 42. Hemoglobin A, Glycosylated/ 43. glyc*mia.mp. 44. insulin*.mp. 45. gly*albumin.mp. 46. OGTT.mp. 47. hba1c.mp. 48. HOMA*.mp. 49. fructosamine*.mp. 50. Insulin/ 51. exp Glucose/ 52. Glucose Tolerance Test/ 53. or/41-52	54. triglyceride.mp. 55. triacylglycerol.mp. 56. VLDL.mp. 57. very low density lipoprotein.mp. 58. lipid*.mp. 59. lipids/ 60. cholesterol/ 61. cholesterol.mp. 62. lipoprotein.mp. 63. lipoproteins/ 64. (hdl or high density lipoprotein).mp. 65. (ldl or low density lipoprotein).mp. 66. exp hyperlipidemias/ 67. apolipoprotein*.mp. 68. non-HDL.mp. 69. or/54-68 70. fatty liver.mp. 71. non-alcoholic fatty liver disease/ 72. NAFLD.mp. 73. transaminases/ 74. alanine transaminase/ 75. alt.mp. 76. aspartate aminotransferase/ 77. ast.mp. 78. IHCL.mp. 79. Intrahepatocellular lipid.mp. 80. transamin*.mp. 81. or/70-80 82. exp uric acid/ 83. uric acid.mp. 84. urate.mp. 85. hyperuricemia/ 86. hyperuricemia.mp. 87. hyperuricaemia.mp. 88. uric.mp. 89. or/82-88 90. "body weight".mp. 91. Exp Body Weight/ 92. Exp Weight Gain/ 93. or/90-92 94. 40 or 53 or 69 or 81 or 89 95. 34 and 94 96. limit 95 to animals 97. 95 not 96 98. review.pt. 99. (medline or medlars or embase or pubmed or Cochrane) 100. (scisearch or psychinfo or psychinfo).tw,sh. 101. (psychlit or psyclit).tw,sh. 102. cinahl.tw,sh. 103. ((hand adj2 search$) or (manual$ adj2 search$)).tw,sh. 104. (electronic database$ or bibliographic database$ or computeri?ed database$ or online database$).tw,sh. 105. (pooling or pooled or mantel haenszel).tw,sh. 106. (peto or dersimonian or der simonian or fixed effect).tw,sh. 107. (retraction of publication or retracted publication).pt. 108. or/99-106	109. 98 and 108 110. meta-analysis.pt. 111. meta-analysis.sh. 112. (meta-analys$ or meta analys$ or metaanalys$).tw,sh. 113. (systematic$ adj5 review$).tw,sh. 114. (systematic$ adj5 overview$).tw,sh. 115. (quantitativ$ adj5 review$).tw,sh. 116. (quantitativ$ adj5 overview$).tw,sh. 117. (quantitativ$ adj5 synthesis$).tw,sh. 118. (methodologic$ adj5 review$).tw,sh. 119. (methodologic$ adj5 overview$).tw,sh. 120. (integrative research review$ or research integration).tw. 121. or/110-119 122. 109 or 121 123. exp cohort studies/ 124. cohort$.tw. 125. controlled clinical trial.pt. 126. epidemiologic methods/ 127. limit 126 to yr=1966-1989 128. exp case-control studies/ 129. (case$ and control$).tw. 130. (case$ and series).tw. 131. case reports.pt. 132. (case$ adj2 report$).tw. 133. (case$ adj2 stud$).tw. 134. or/123-125,127-132 135. 122 or 134 136. 97 not 135 137. randomized controlled trial.pt. 138. (random$ or placebo$ or single blind$ or double blind$ or triple blind$).ti,ab. 139. (retraction of publication or retracted publication).pt. 140. or/137-139 141. (animals not humans).sh. 142. ((comment or editorial or meta-analysis or practice-guideline or review or letter) not randomized controlled trial).pt. 143. (random sampl$ or random digit$ or random effect$ or random survey or random regression).ti,ab. not randomized controlled trial.pt. 144. or/141-143 145. 140 not 144 146. 136 and 145 147. 97 and 145 148. from 147 keep 1-3511

[1] For all databases, the original search date was 28 March 2019, updated 31 March 2021 and then finally on 26 March 2021

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eTable 2. Search strategy EMBASE

Database	EMBASE, Cochrane[1]		
Search Terms	1. aspartame.mp. 2. exp Aspartame/ 3. neotame.mp. 4. saccharin.mp. 5. exp Saccharin/ 6. sucralose.mp. 7. stevia.mp. 8. exp Stevia/ 9. acesulfame.mp. 10. Sweetening Agents/ 11. sugar substitute*.mp. 12. noncaloric.mp. 13. non-caloric.mp. 14. nonnutritive.mp. 15. non-nutritive.mp. 16. no calorie*.mp. 17. low calorie sweeten*.mp. 18. sugar-free.mp. 19. artificial sweet*.mp. 20. diet beverage*.mp. 21. diet soda.mp. 22. exp Carbonated Beverages/ 23. ssb.mp. 24. sugar*sweetened beverage*.mp. 25. Fructose/ 26. fructose.mp. 27. dietary sucrose/ 28. sucrose.mp. 29. High Fructose Corn Syrup/ 30. High Fructose Corn Syrup.mp. 31. Soft drink*.mp. 32. cola.mp. 33. sugar*.mp. 34. or/1-33 35. Blood Pressure/ 36. systolic blood pressure.mp. 37. SBP.mp. 38. diastolic blood pressure.mp. 39. DBP.mp. 40. or/35-39 41. glyc*m*.mp. 42. Hemoglobin A, Glycosylated/ 43. glyc*mia.mp. 44. insulin*.mp. 45. gly*albumin.mp. 46. OGTT.mp. 47. hba1c.mp. 48. HOMA*.mp. 49. fructosamine*.mp. 50. Insulin/ 51. exp Glucose/ 52. Glucose Tolerance Test/ 53. or/41-52	54. triglyceride.mp. 55. triacylglycerol.mp. 56. VLDL.mp. 57. very low density lipoprotein.mp. 58. lipid*.mp. 59. lipids/ 60. cholesterol/ 61. cholesterol.mp. 62. lipoprotein.mp. 63. lipoproteins/ 64. (hdl or high density lipoprotein).mp. 65. (ldl or low density lipoprotein).mp. 66. exp hyperlipidemias/ 67. apolipoprotein*.mp. 68. non-HDL.mp. 69. or/54-68 70. fatty liver.mp. 71. non-alcoholic fatty liver disease/ 72. NAFLD.mp. 73. transaminases/ 74. alanine transaminase/ 75. alt.mp. 76. aspartate aminotransferase/ 77. ast.mp. 78. IHCL.mp. 79. Intrahepatocellular lipid.mp. 80. transamin*.mp. 81. or/70-80 82. exp uric acid/ 83. uric acid.mp. 84. urate.mp. 85. hyperuricemia/ 86. hyperuricemia.mp. 87. hyperuricaemia.mp. 88. uric.mp. 89. or/82-88 90. “body weight”.mp. 91. Exp Body Weight/ 92. Exp Weight Gain/ 93. or/90-92 94. 40 or 53 or 69 or 81 or 89 or 93 95. 34 and 94 96. limit 95 to animals 97. 95 not 96 98. exp review/ 99. (literature adj3 review$).ti,ab. 100. exp meta analysis/ 101. exp Systematic Review/ 102. or/98-101 103. (medline or medlars or embase or pubmed or cinahl or amed or psychlit or psyclit or psychinfo or psycinfo or scisearch or cochrane).ti,ab. 104. RETRACTED ARTICLE/ 105. 103 or 104 106. 102 or 105 107. (systematic$ adj2 (review$ or overview)).ti,ab. 108. o(meta?anal$ or meta anal$ or meta-anal$ or metaanal$ or metanal$).ti,ab. 109. or/106-108	110. exp cohort analysis/ 111. exp longitudinal study/ 112. exp prospective study/ 113. exp follow up/ 114. cohort$.tw. 115. exp case control study/ 116. (case$ and control$).tw. 117. exp case study/ 118. (case$ and series).tw. 119. case report/ 120. (case$ adj2 report$).tw. 121. (case$ adj2 stud$).tw. 122. or/110-121 123. 106 or 122 124. 97 not 123 125. (random$ or placebo$ or single blind$ or double blind$ or triple blind$).ti,ab. 126. RETRACTED ARTICLE/ 127. 125 or 126 128. (animal$ not human$).sh,hw. 129. (book or conference paper or editorial or letter or review).pt. not exp randomized controlled trial/ 130. (random sampl$ or random digit$ or random effect$ or random survey or random regression).ti,ab. not exp randomized controlled trial/ 131. or/128-130 132. 127 not 131 133. 124 and 132

[1] For all databases, the original search date was 28 March 2019, updated 31 March 2021 and then finally on 26 March 2021

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eTable 3. Search strategy Cochrane

Database	Cochrane[1]	
Search Terms	1. aspartame.mp. 2. exp Aspartame/ 3. neotame.mp. 4. saccharin.mp. 5. exp Saccharin/ 6. sucralose.mp. 7. stevia.mp. 8. exp Stevia/ 9. acesulfame.mp. 10. Sweetening Agents/ 11. sugar substitute*.mp. 12. noncaloric.mp. 13. non-caloric.mp. 14. nonnutritive.mp. 15. non-nutritive.mp. 16. low calorie sweeten*.mp. 17. sugar-free.mp. 18. artificial sweet*.mp. 19. diet beverage*.mp. 20. diet soda.mp. 21. exp Carbonated Beverages/ 22. ssb.mp. 23. sugar*sweetened beverage*.mp. 24. Fructose/ 25. fructose.mp. 26. dietary sucrose/ 27. sucrose.mp. 28. High Fructose Corn Syrup/ 29. High Fructose Corn Syrup.mp. 30. Soft drink*.mp. 31. cola.mp. 32. sugar*.mp. 33. or/1-32 34. Blood Pressure/ 35. systolic blood pressure.mp. 36. SBP.mp. 37. diastolic blood pressure.mp. 38. DBP.mp. 39. Or/35-39 40. glyc*m*.mp. 41. Hemoglobin A, Glycosylated/ 42. glyc*mia.mp. 43. insulin*.mp. 44. gly*albumin.mp. 45. OGTT.mp. 46. hba1c.mp. 47. HOMA*.mp. 48. fructosamine*.mp. 49. Insulin/ 50. exp Glucose/ 51. Glucose Tolerance Test/ 52. or/41-52	53. triglyceride.mp. 54. triacylglycerol.mp. 55. VLDL.mp. 56. very low density lipoprotein.mp. 57. lipid*.mp. 58. lipids/ 59. cholesterol/ 60. cholesterol.mp. 61. lipoprotein.mp. 62. lipoproteins/ 63. (hdl or high density lipoprotein).mp. 64. (ldl or low density lipoprotein).mp. 65. exp hyperlipidemias/ 66. apolipoprotein*.mp. 67. non-HDL.mp. 68. or/54-68 69. fatty liver.mp. 70. non-alcoholic fatty liver disease/ 71. NAFLD.mp. 72. transaminases/ 73. alanine transaminase/ 74. alt.mp. 75. aspartate aminotransferase/ 76. ast.mp. 77. IHCL.mp. 78. Intrahepatocellular lipid.mp. 79. transamin*.mp. 80. or/70-80 81. exp uric acid/ 82. uric acid.mp. 83. urate.mp. 84. hyperuricemia/ 85. hyperuricemia.mp. 86. hyperuricaemia.mp. 87. uric.mp. 88. or/82-88 89. “body weight”.mp. 90. exp Body Weight/ 91. exp Weight Gain/ 92. or/90-92 93. 39 or 52 or 68 or 80 or 88 or 92 94. 33 and 93 95. limit 95 to randomized controlled trial

[1] For all databases, the original search date was 28 March 2019, updated 31 March 2021 and then finally on 26 March 2021

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eTable 4. PICOTS[b] framework

Participants	Intervention	Comparison	Outcome	Time	Study Design
Adult men and women excluding pregnant or breastfeeding women and children. All health backgrounds	LNCSBs or SSBs or water	LNCSBs or SSBs or water	Adiposity, glycemic control, established blood lipid targets, blood pressure, non-alcoholic fatty liver and uric acid	≥ 2 weeks	Human randomized controlled trials

[b]Hutton B, Salanti G, Caldwell DM, Chaimani A, Schmid CH, Cameron C, Ioannidis JP, Straus S, Thorlund K, Jansen JP, Mulrow C, Catalá-López F, Gøtzsche PC, Dickersin K, Boutron I, Altman DG, Moher D. The PRISMA extension statement for reporting of systematic reviews incorporating network meta-analyses of health care interventions: checklist and explanations. Ann Intern Med. 2015 Jun 2;162(11):777-84.

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eTable 5. Trial characteristics

Study	Subjects	Age Mean (SD)	Sex No. (%) Males:Females	LNCS Type	Beverage Dose (mL/d)			Setting	Design	Duration (weeks)	Funding Source	Contacted for data / Provided data
					LNCSB	Water	SSB					
Bonnet et al. 2018	50	31.1 (10.3)	22:28	Asp/ Ace-K	660	660	NA	France	C	12	Agency/ Industry	Yes / No
LNCSB:Water	(22M:28F)		(44:56)									
*Bruun et al. 2015	35	39 (1.1)	14:21	Asp	1000	1000	1000	Denmark	P	26	Agency/ Industry	Yes / No
LNCSB	(3M:9F)	39 (27.7)	(40:60)									
Water	(5M:8F)	39 (28.8)										
SSB	(6M:4F)	39 (19)										
Ebbeling et al. 2020		26.8(5.6)	121:82	NR	355	355	355	USA	P	52	Agency	
LNCSB	67 (40M, 27F)	26.7(5.7)	(60:40)									
Water	69 (41M, 28F)	27.9(6.0)										
SSB	67 (40M, 27F)	25.9(5.1)										
Campos et al. 2015	27	NR	14:13	NR	1300	NA	1300	Switzerland	P	12	Agency	Yes / Yes
LNCSB	(8M:6F)		(52:48)									
SSB	(6M:7F)											
*Engel et al. 2017	45	38.6 (7.6)	16:29	NR	1000	1000	1000	Denmark	P	26	Agency/ Industry	Yes / No
LNCSB	(3M:12F)	39 (7.6)	(36:64)									
Water	(5M:11F)	39 (7.3)										
SSB	(8M:6F)	37.8 (8)										
Hernández-Cordero et al. 2014	240	33.3 (6.7)	0:240 (0:100)	NR	NA	250+	250+	Mexico	P	39	Industry	
Water	120F	33.5 (6.7)										
SSB	120F	33.3 (6.7)										
Higgins et al. 2018	93	22.9 (1.0)	43:50	Asp	500	500	NA	USA	P	12	Industry	
Water	(13M:18F)	24.2 (1.3)	(46:54)									
350mg LNCSB	(17M:14F)	22.8 (1.0)										
1050mg LNCSB	(13M:18F)	21.8 (0.6)										
Higgins & Mattes 2019	154	27.3 (9.6)	67:87 (44:56)	Sac, Asp, RebA, Suc	1250-1750	NA	1250-1750	USA	P	12	Agency	Yes / Yes
LNCSB (Sac)	(12M:17F)	25.8 (6.9)										
LNCSB (Asp)	(15M:15F)	29.5 (12)										
LNCSB (RebA)	(10M:18F)	27.1 (9.6)										
LNCSB (Suc)	(12M:16F)	25.9 (9)										
SSB	(18M:21F)	28.2 (9.5)										
Madjd et al. 2015	62	32 (6.9)	0:62 (0:100)	NR	250+	250+	NA	Iran	P	24	Agency	
LNCSB	32F	31.7 (6.8)										
Water	30F	32.2 (6.9)										
Madjd et al. 2017	81	34.8 (7.2)	0:81 (0:100)	NR	250+	250+	NA	Iran	P	24	Agency	
LNCSB	40F	35.5 (7.5)										
Water	41F	34.2 (7)										
Maersk et al. 2012	35	39 (26)	14:21 (40:60)	NR	1000	1000	1000	Denmark	P	26	Agency/ Industry	
LNCSB	(3M:9F)	39(27.7)										
Water	(5M:8F)	39(28.8)										
SSB	(6M:4F)	39(19)										
Peters et al. 2016	308	47.8 (10.5)	53:255 (17:83)	NR	710	710	NA	USA	P	52	Industry	
LNCSB	(28M:130F)	48.3 (10.4)										
Water	(25M:125F)	47.3 (10.6)										
Reid et al. 2007	133	31.8 (9.1)	0:133 (0:100)	Asp	1000	NA	1000	England	P	4	Agency	
LNCSB	65F	NR										
SSB	68F	NR										
Reid et al. 2010	53	33.7 (9.9)	0:53 (0:100)	Asp	1000	NA	1000	Scotland	P	4	Agency	Yes / No
LNCSB	29F	32.9 (8.8)										
SSB	24F	34.5 (11)										
Reid et al. 2014	41	35 (9.1)	0:41 (0:100)	Asp	1000	NA	1000	Scotland	P	4	Agency/ Industry	Yes / No
LNCSB	21F	34.6 (8.5)										
SSB	20F	35.1 (9.9)										
Tate et al. 2012	213	42 (10.7)	35:178 (52:48)	NR	1420-2000	1420- 2000	NA	USA	P	26	Industry	
LNCSB	(23M:82F)	41.2 (11.2)										
Water	(12M:96F)	43.2 (10.6)										
Tordoff et al. 1990	30	25.6 (5.3)	21:9 (70:30)	Asp	1135	NA	1135	USA	C	3	Agency	
LNCSB: SSB	(21M:9F)											

Asp, aspartame; Ace-K, acesulfame potassium; NA, not applicable; No., number; LNCS, low and non-nutritive sweetener; LNCSB, low and non-nutritive sweetened beverage; NW, normal weight; NR, not reported; OB, obese; OW, overweight; RebA, rebaudioside A; Sac, saccharin; SD, standard deviation; Suc, sucralose; SSB, sugar-sweetened beverage; T2DM, type 2 diabetes mellitus.
*Secondary analyses to Maersk et al. 2012. As more outcomes were reported in the Engel et al. 2017 analysis, data from that trial was used for the majority of outcomes, except for uric acid (Bruun et al.) and IHCL (Maersk et al).

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eTable 6. Loop-specific approach for inconsistency

Outcome	P-value	Inconsistency
Body weight	P=0.074	Not present
BMI	P=0.8	Not present
Body fat	P=0.98	Not present
Waist circumference	Unable to calculate*	
HbA1c	Unable to calculate*	
FPG	P=0.25	Not present
2h-PG	No triangular loop found	
FPI	P=0.75	Not present
HOMA-IR	P=0.76	Not present
LDL-C	P=0.95	Not present
Non-HDL-C	P=0.66	Not present
TGs	P=0.76	Not present
HDL-C	P=0.49	Not present
TC	P=0.63	Not present
SBP	P=0.15	Not present
DBP	P=0.57	Not present
IHCL	Unable to calculate*	
ALT	Unable to calculate*	
AST	Unable to calculate*	
Uric acid	Unable to calculate*	

2h-PG, two-hour post prandial glucose; ALT, alanine transaminase; AST, aspartate transaminase; BMI, body mass index; DBP, diastolic blood pressure; FPG, fasting plasma glucose; FPI, fasting plasma insulin; HbA1c, hemoglobin A1c; HDL-C, high-density lipoprotein-cholesterol; HOMA-IR, Homeostatic Model Assessment of Insulin Resistance; IHCL, intra-hepatocellular lipid; LDL-C, low-density lipoprotein-cholesterol; Non-HDL-C, non-high-density lipoprotein-cholesterol; SBP, systolic blood pressure; TC, total cholesterol; TGs, triglycerides.

* Calculation issues with loop-specific approach in Stata with multiple-arm trials as model did not converge, therefore, “unable to calculate” for all missing ones.

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eTable 7. Design by treatment approach for inconsistency

Outcome	P-value	Inconsistency
Body weight	0.99	Not present
BMI	0.71	Not present
Body fat	0.94	Not present
Waist circumference	0.88	Not present
HbA1c	0.83	Not present
FPG	0.85	Not present
2h-PG	0.06	Not present
FPI	0.27	Not present
HOMA-IR	0.45	Not present
LDL-C	0.28	Not present
Non-HDL-C	0.075	Not present
TGs	0.45	Not present
HDL-C	0.77	Not present
TC	0.08	Not present
SBP	0.77	Not present
DBP	0.27	Not present
IHCL	Unable to calculate*	Unable to assess*
ALT	0.51	Not present
AST	0.92	Not present
Uric acid	0.76	Not present

2h-PG, two-hour post prandial glucose; ALT, alanine transaminase; AST, aspartate transaminase; BMI, body mass index; DBP, diastolic blood pressure; FPG, fasting plasma glucose; FPI, fasting plasma insulin; HbA1c, hemoglobin A1c; HDL-C, high-density lipoprotein-cholesterol; HOMA-IR, Homeostatic Model Assessment of Insulin Resistance; IHCL, intra-hepatocellular lipid; LDL-C, low-density lipoprotein-cholesterol; Non-HDL-C, non-high-density lipoprotein-cholesterol; SBP, systolic blood pressure; TC, total cholesterol; TGs, triglycerides.
*Calculation issues with design-by-treatment inconsistency in Stata did not converge.

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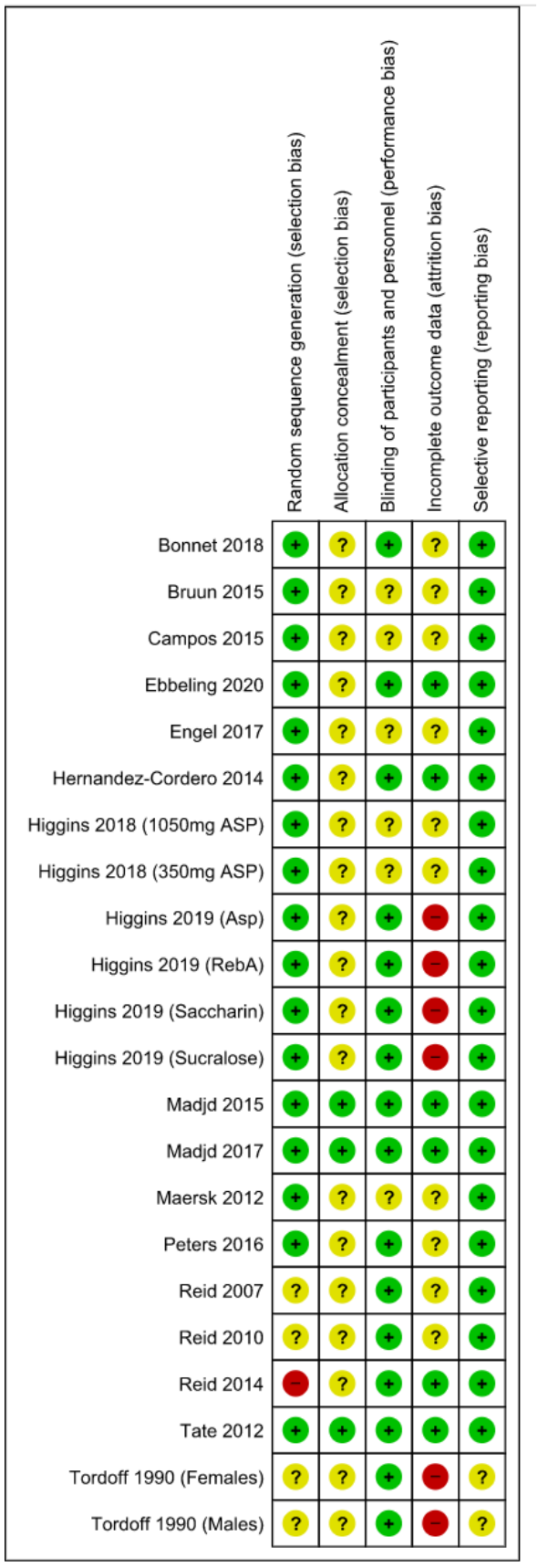
eFigures

eFigure 1. Cochrane risk of bias summary for all included trials



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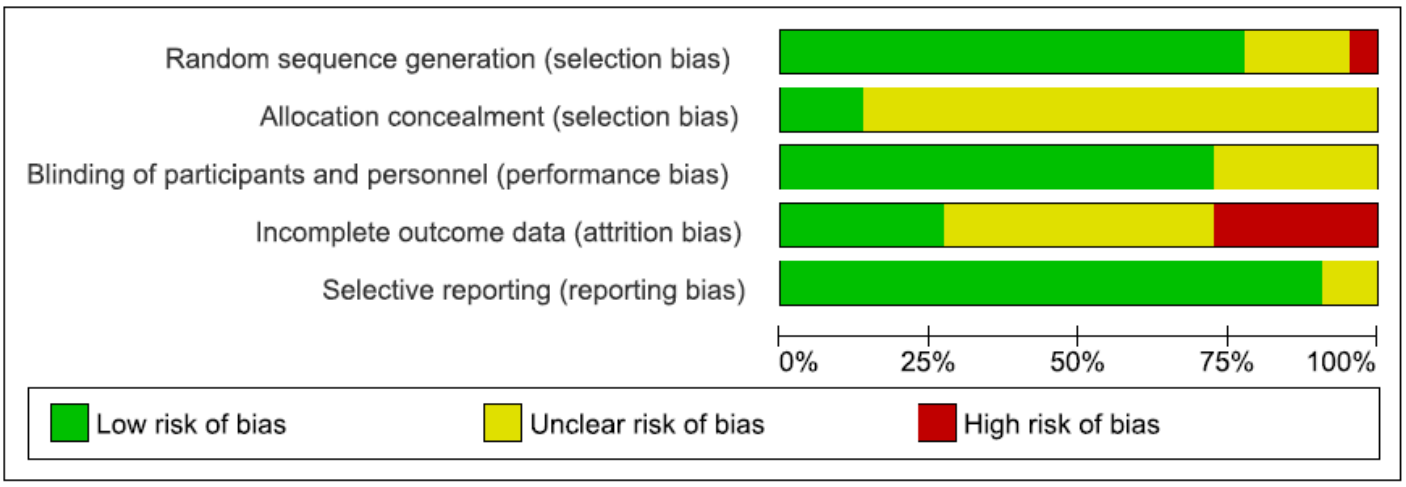
eFigure 2. Risk of bias proportion for all included trials



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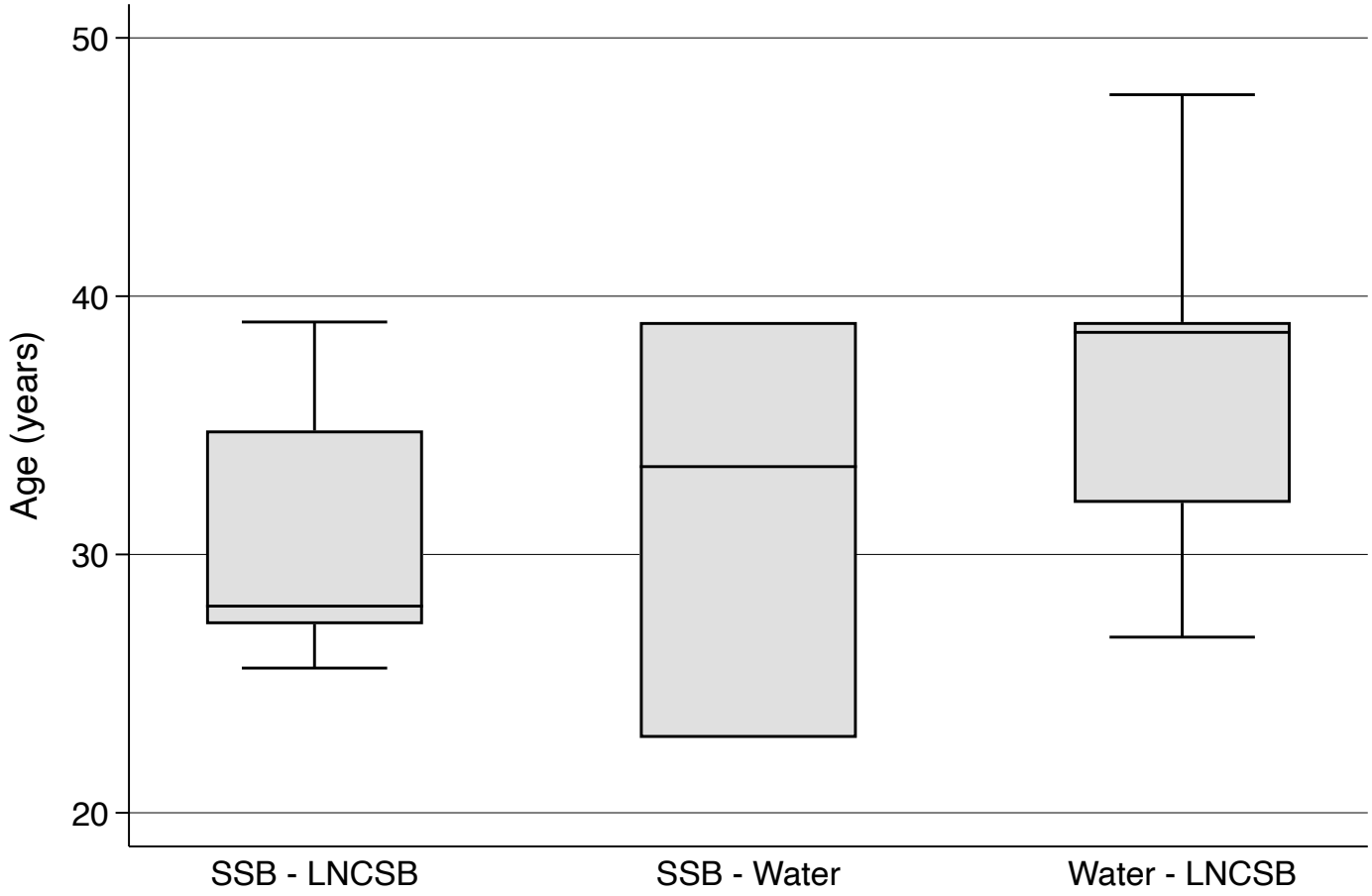
eFigure 3. Transitivity analysis box plots showing the distribution of the mean age (years) of the trials across the available direct comparisons



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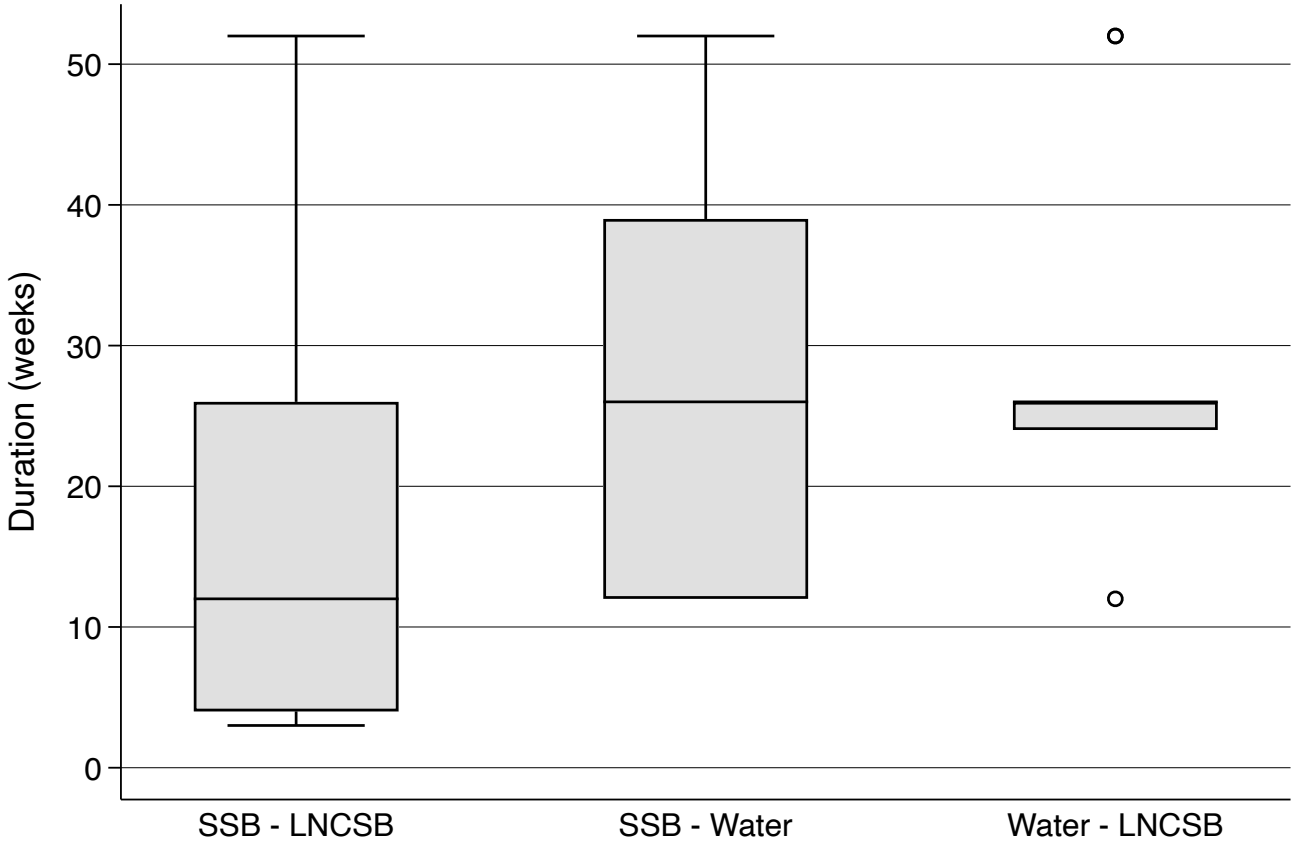
eFigure 4. Transitivity analysis box plots showing the distribution of the study length (weeks) of the trials across the available direct comparisons



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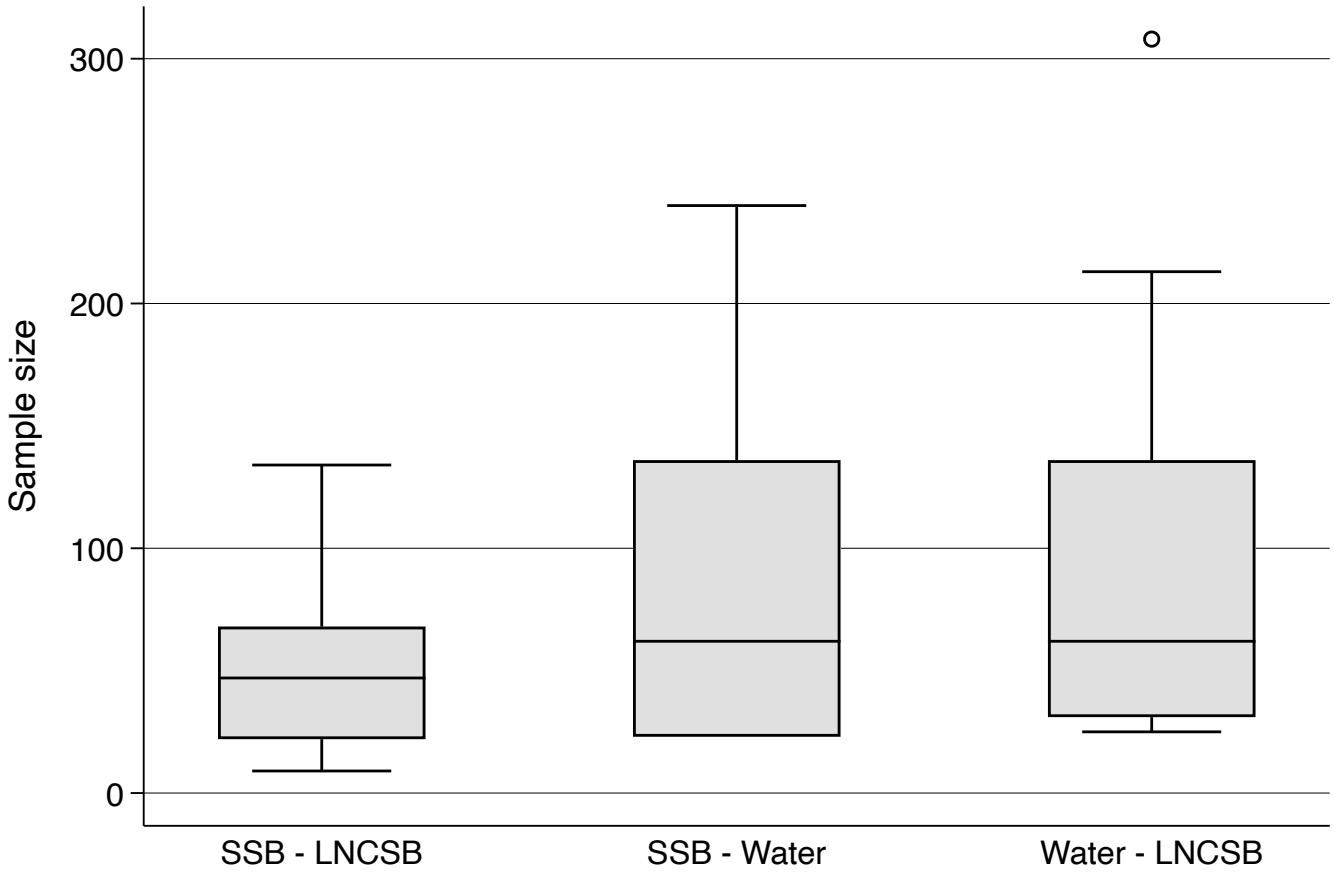
eFigure 5. Transitivity analysis box plots showing the distribution of the sample size of the trials across the available direct comparisons



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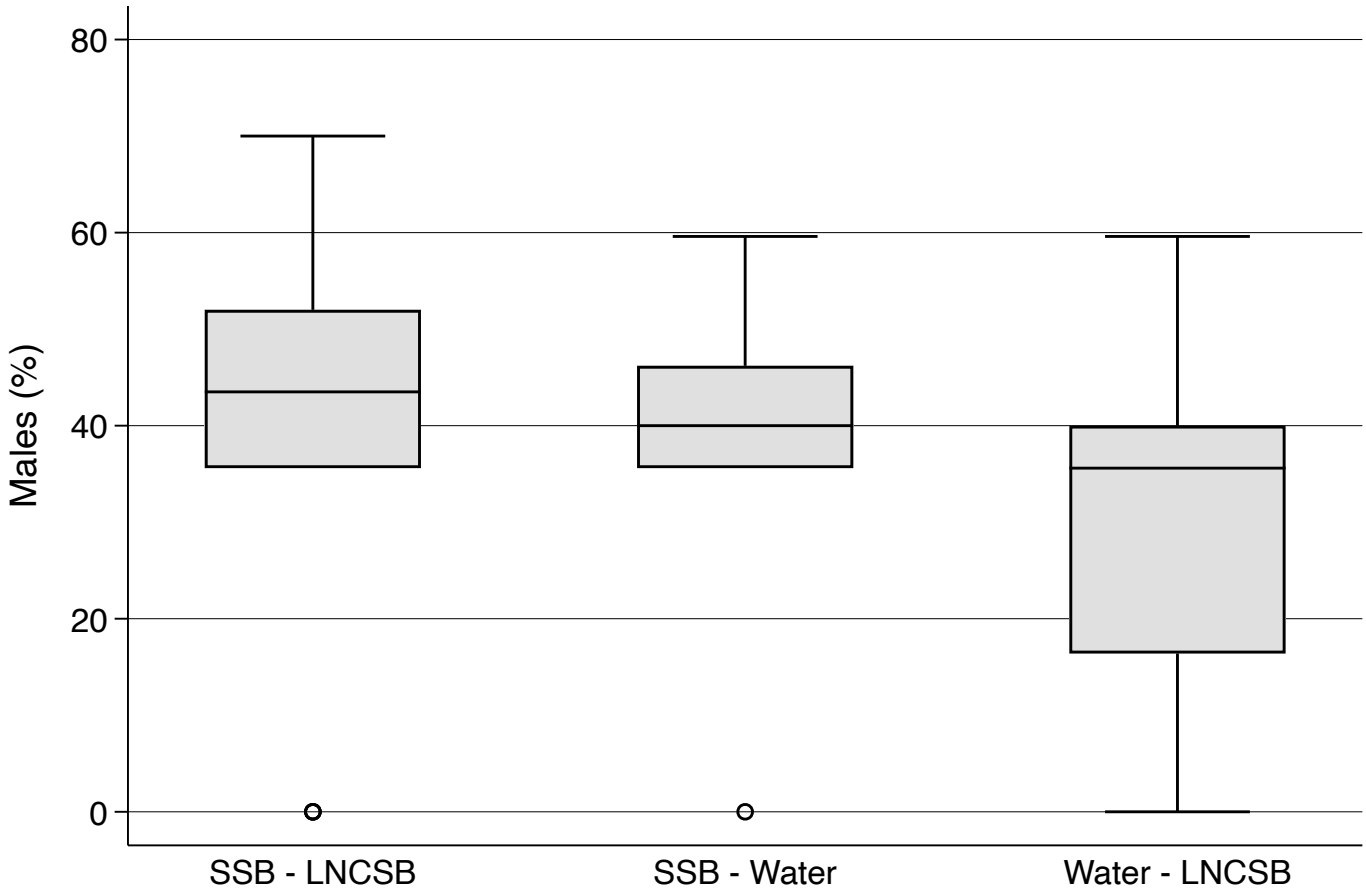
eFigure 6. Transitivity analysis box plots showing the distribution of % males of the trials across the available direct comparisons



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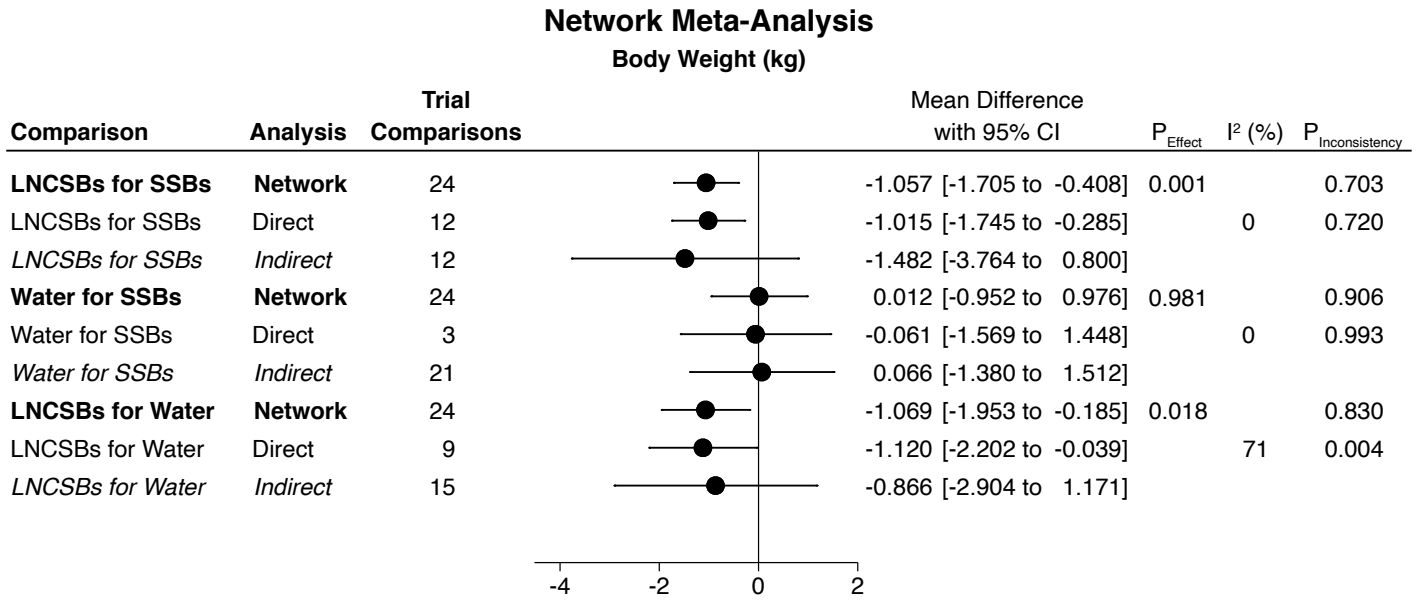
eFigure 7. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on Body Weight



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication bias	Overall
LNCSBs for SSBs	Not serious	Not serious	Not serious	Serious *Rated down the evidence for serious imprecision in the network estimates as the 95% CIs were wide and crossed prespecified MIDs for benefit and/or harm (0.5 kg for body weight).*	Not serious	**Moderate**
Water for SSBs	Not serious	Serious *Rated down the evidence for incoherence on visual inspection.*	Not serious	Serious *Rated down the evidence for serious imprecision in the network estimates as the 95% CIs were wide and crossed prespecified MIDs for benefit and/or harm (0.5 kg for body weight).*	Not serious	**Low**
LNCSBs for Water	Not serious	Serious *Rated down for inconsistency assessed in the direct estimates as substantial unexplained heterogeneity (I2 >50%, P<0.10)*	Not serious	Serious *Rated down the evidence for serious imprecision in the network estimates as the 95% CIs were wide and crossed prespecified MIDs for benefit and/or harm (0.5 kg for body weight).*	Not serious	**Low**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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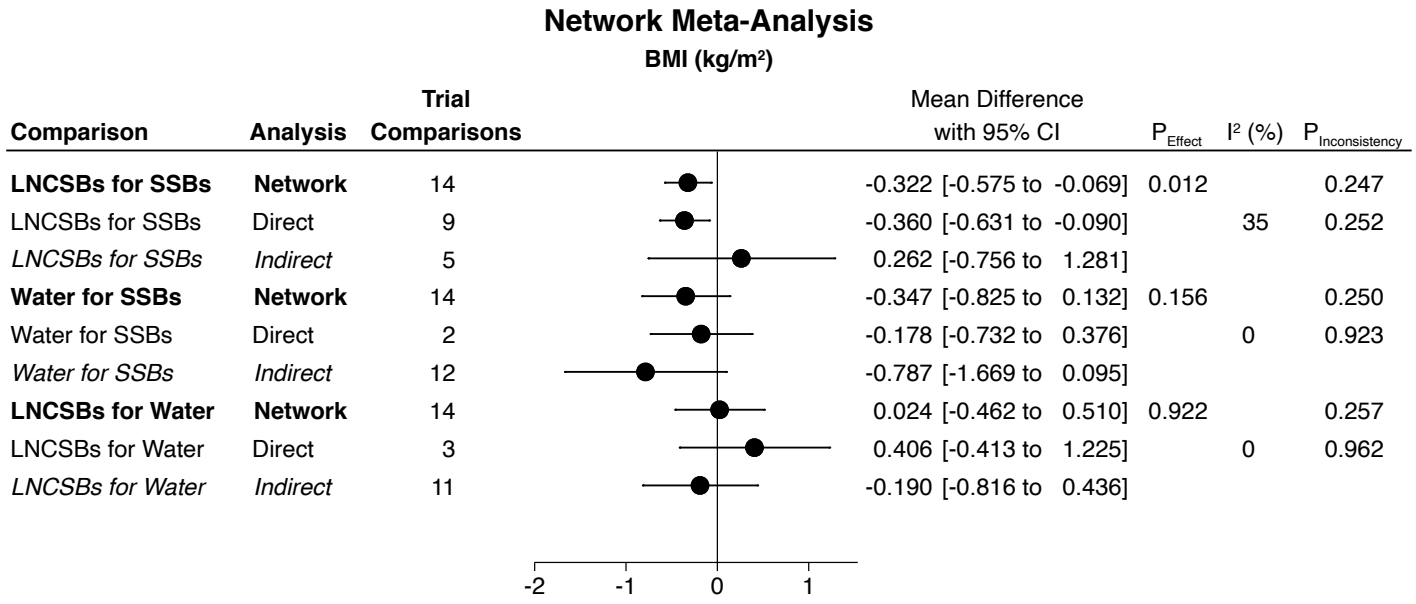
eFigure 8. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on BMI



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication bias	Overall
LNCSBs for SSBs	Not serious	Serious *Rated down the evidence for incoherence via visual inspection.*	Not serious	Serious *Rated down the evidence for serious imprecision in the network estimates as the 95% CIs were wide and crossed prespecified MIDs for benefit and/or harm (0.2kg/m2 for BMI)*	Not serious	**Low**
Water for SSBs	Not serious	Not serious	Not serious	Serious *Rated down the evidence for serious imprecision in the network estimates as the 95% CIs were wide and crossed prespecified MIDs for benefit and/or harm (0.2kg/m2 for BMI)*	Not serious	**Moderate**
LNCSBs for Water	Not serious	Serious *Rated down the evidence for incoherence on visual inspection.*	Not serious	Serious *Rated down the evidence for serious imprecision in the network estimates as the 95% CIs were wide and crossed prespecified MIDs for benefit and/or harm (0.2kg/m2 for BMI)*	Not serious	**Low**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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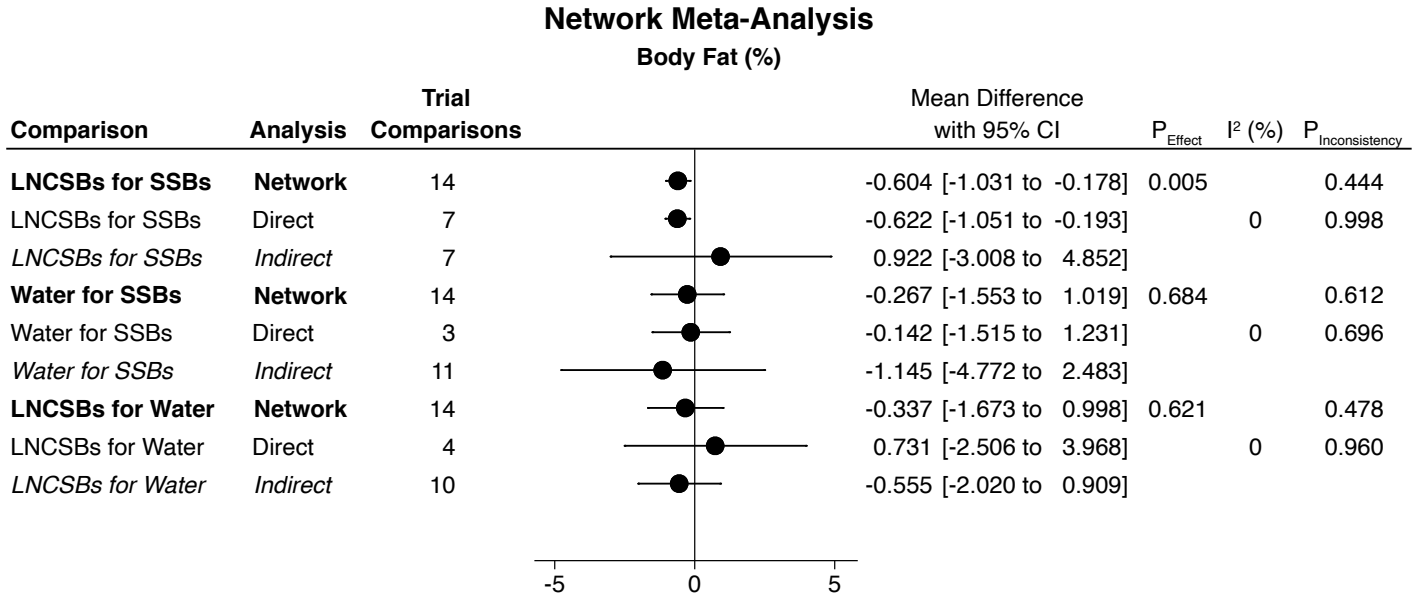
eFigure 9. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on body fat (%)



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication bias	Overall
LNCSBs for SSBs	Not serious	Serious *Rated down the evidence for incoherence via visual inspection.*	Not serious	Not serious *(MIDs for body fat = 2%)*	Not serious	**MODERATE**
Water for SSBs	Not serious	Not serious	Not serious	Not serious *(MIDs for body fat = 2%)*	Not serious	**HIGH**
LNCSBs for Water	Not serious	Serious *Rated down the evidence for incoherence via visual inspection.*	Not serious	Not serious *(MIDs for body fat = 2%)*	Not serious	**MODERATE**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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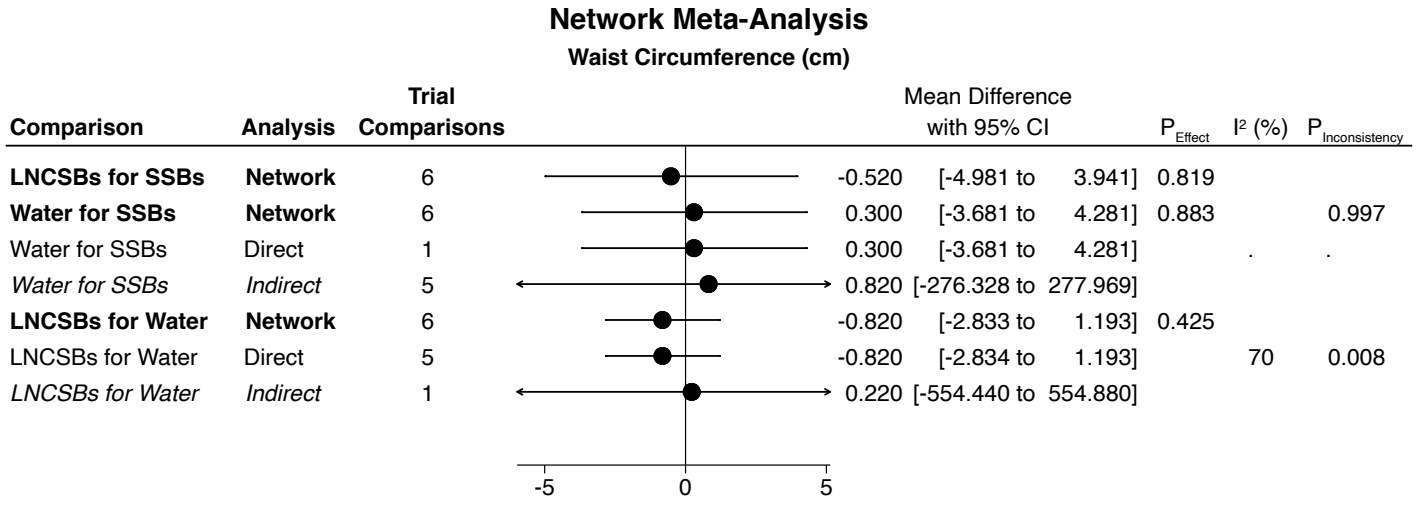
eFigure 10. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on waist circumference



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication bias	Overall
LNCSBs for SSBs	Not serious	Not serious	Serious *Rated down for indirectness as only based upon indirect assessment.*	Serious *Rated down the evidence for serious imprecision in the network estimates as the 95% CIs were wide and crossed prespecified MIDs for benefit and/or harm. MIDs for waist circumference = 2 cm).*	**NA**	**LOW**
Water for SSBs	Not serious	Not serious	Serious *Rated down for indirectness as only 1 direct study available.*	Serious *Rated down the evidence for serious imprecision in the network estimates as the 95% CIs were wide and crossed prespecified MIDs for benefit and/or harm. MIDs for waist circumference = 2 cm).*	**NA**	**LOW**
LNCSBs for Water	Not serious	Serious *Rated down for inconsistency assessed in the direct estimates as substantial unexplained heterogeneity (I2 >50%, P<0.10)*	Not serious.	Serious *Rated down the evidence for serious imprecision in the network estimates as the 95% CIs were wide and crossed prespecified MIDs for benefit and/or harm. MIDs for waist circumference = 2 cm).*	**NA**	**LOW**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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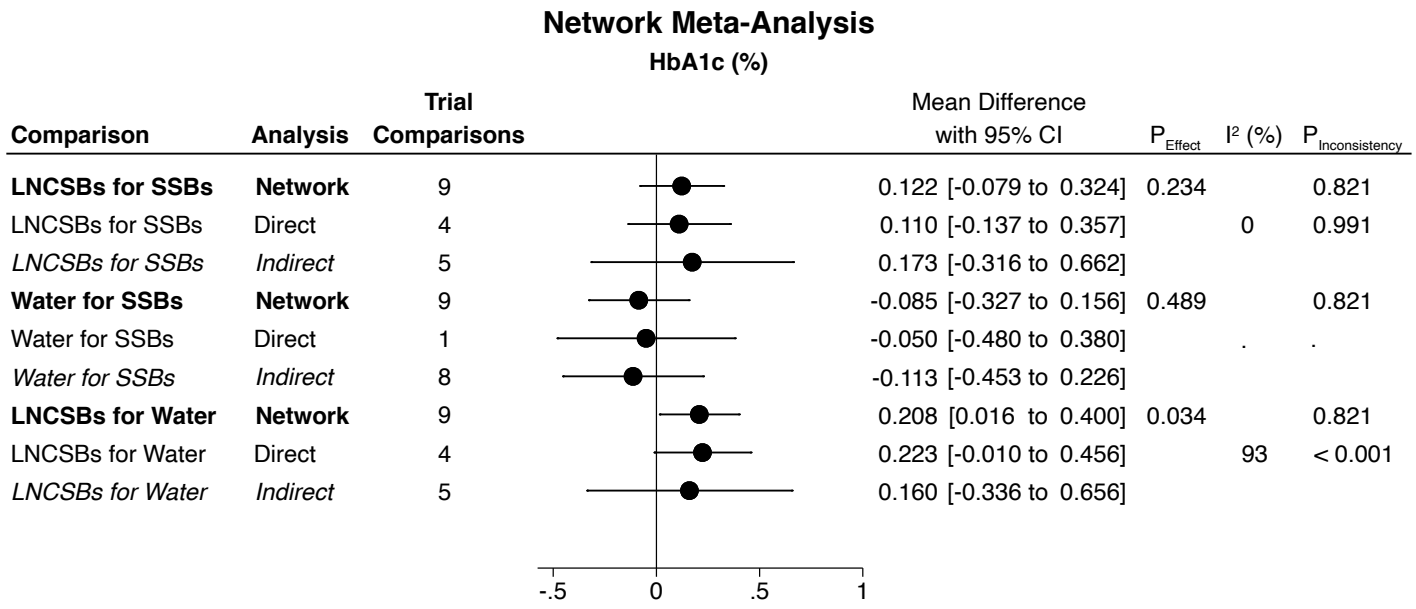
eFigure 11. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on HbA1c



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall
LNCSBs for SSBs	Not serious	Not serious	Not serious	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MID = 0.3% for HbA1c),*	NA	**MODERATE**
Water for SSBs	Not serious	Not serious	Serious *Rated down for indirectness as only 1 direct study available.*	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MID = 0.3% for HbA1c),*	NA	**LOW**
LNCSBs for Water	Not serious	Serious *Rated down for inconsistency assessed in the direct estimates as substantial unexplained heterogeneity (I2 >50%, P<0.10)*	Not serious	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MID = 0.3% for HbA1c),*	NA	**LOW**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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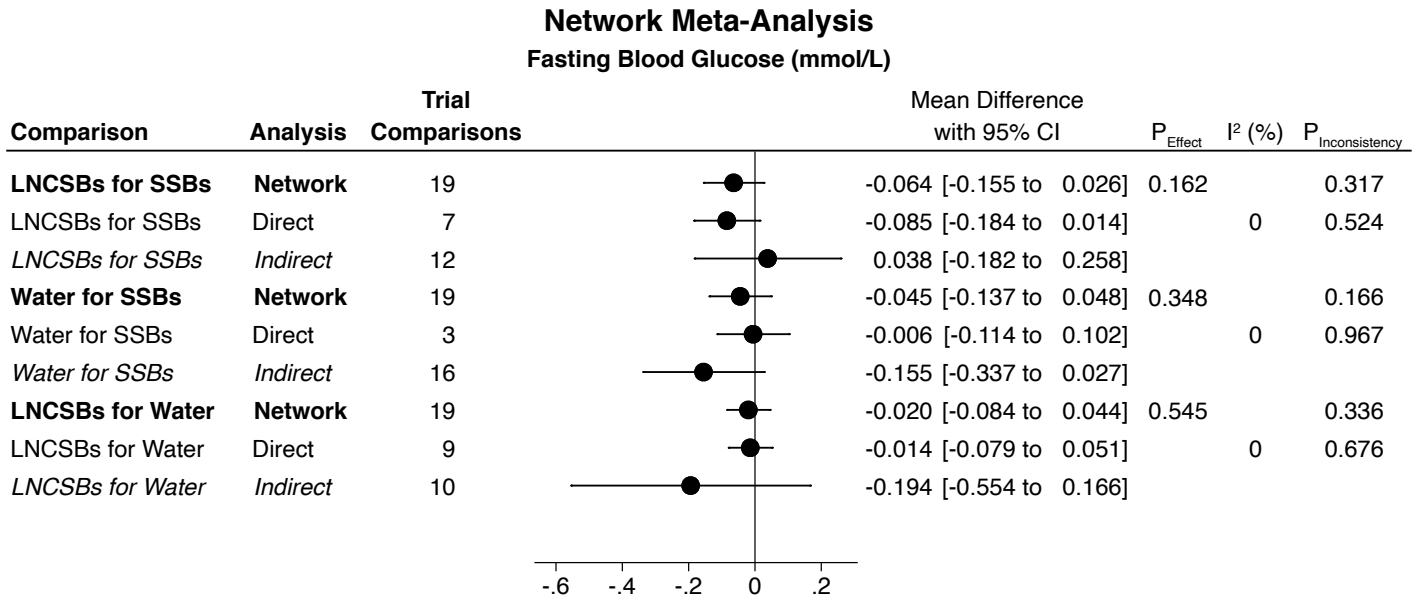
eFigure 12. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on fasting plasma glucose (FPG)



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall
LNCSBs for SSBs	Not serious	Serious *Rated down the evidence for incoherence via visual inspection.*	Not serious	Not serious *(MID for FPG = 0.5 mmol/L)*	Not serious	**MODERATE**
Water for SSBs	Not serious	Not serious	Not serious	Not serious *(MID for FPG = 0.5 mmol/L)*	Not serious	**HIGH**
LNCSBs for Water	Not serious	Not serious	Not serious	Not serious *(MID for FPG = 0.5 mmol/L)*	Not serious	**HIGH**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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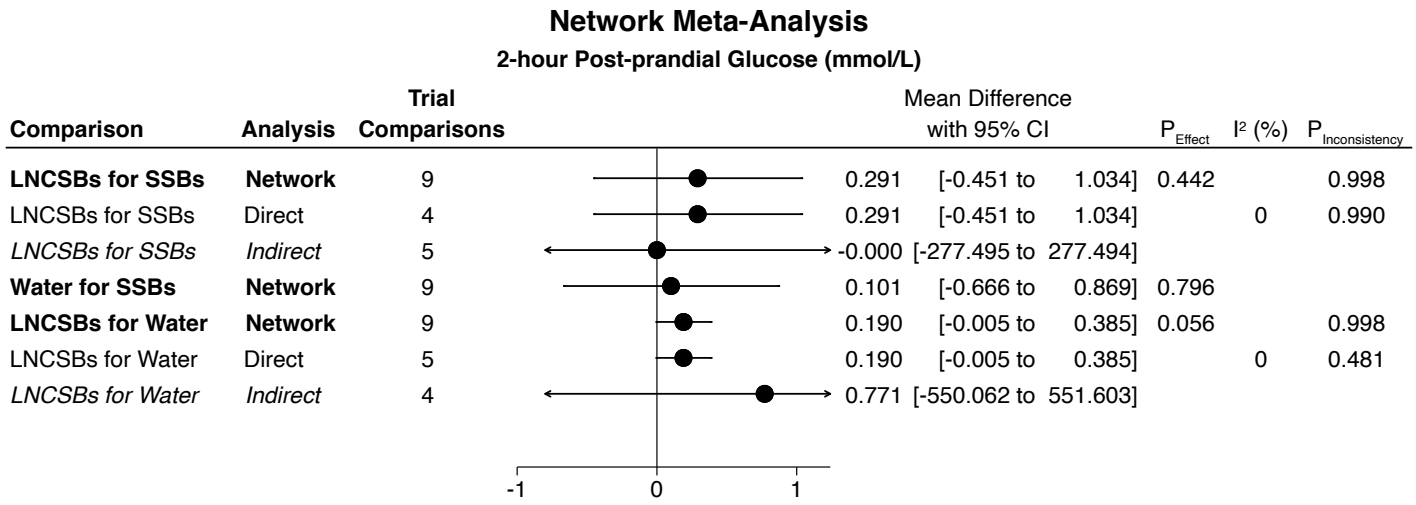
eFigure 13. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on 2-hour post-prandial glucose (2HPP)



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall
LNCSBs for SSBs	Not serious	Not serious	Not serious	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for 2h-PG = 0.1%),*	NA	**MODERATE**
Water for SSBs	Not serious	Not serious	Serious *Rated down for indirectness as only based upon indirect assessment.*	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for 2h-PG = 0.1%),*	NA	**LOW**
LNCSBs for Water	Not serious	Not serious	Not serious	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for 2h-PG = 0.1%),*	NA	**MODERATE**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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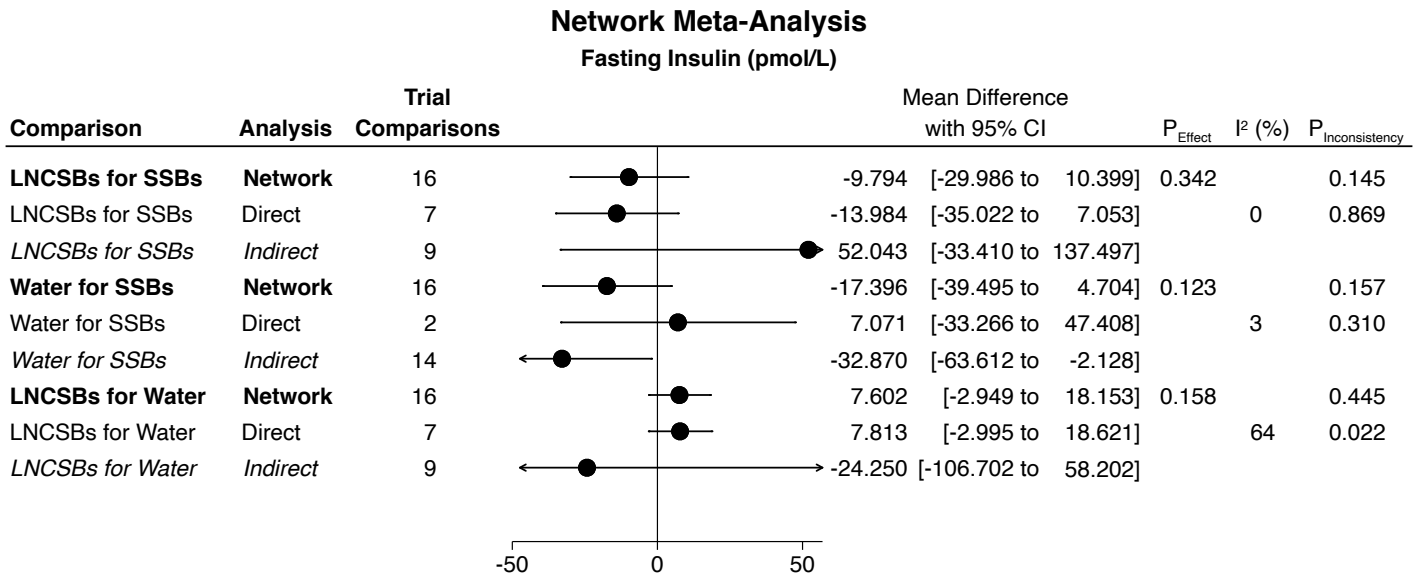
eFigure 14. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on fasting insulin



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall
LNCSBs for SSBs	Not serious	Serious *Rated down the evidence for incoherence via visual inspection.*	Not serious.	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for FPI = 5pmol)*	Not serious	**LOW**
Water for SSBs	Not serious	Not serious *Rated down for incoherence via visual inspection.*	Not serious.	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for FPI = 5pmol)*	Not serious	**LOW**
LNCSBs for Water	Not serious	Serious *Rated down for inconsistency assessed in the direct estimates as substantial unexplained heterogeneity (I2 >50%, P<0.10)*	Not serious.	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for FPI = 5pmol)*	Not serious	**LOW**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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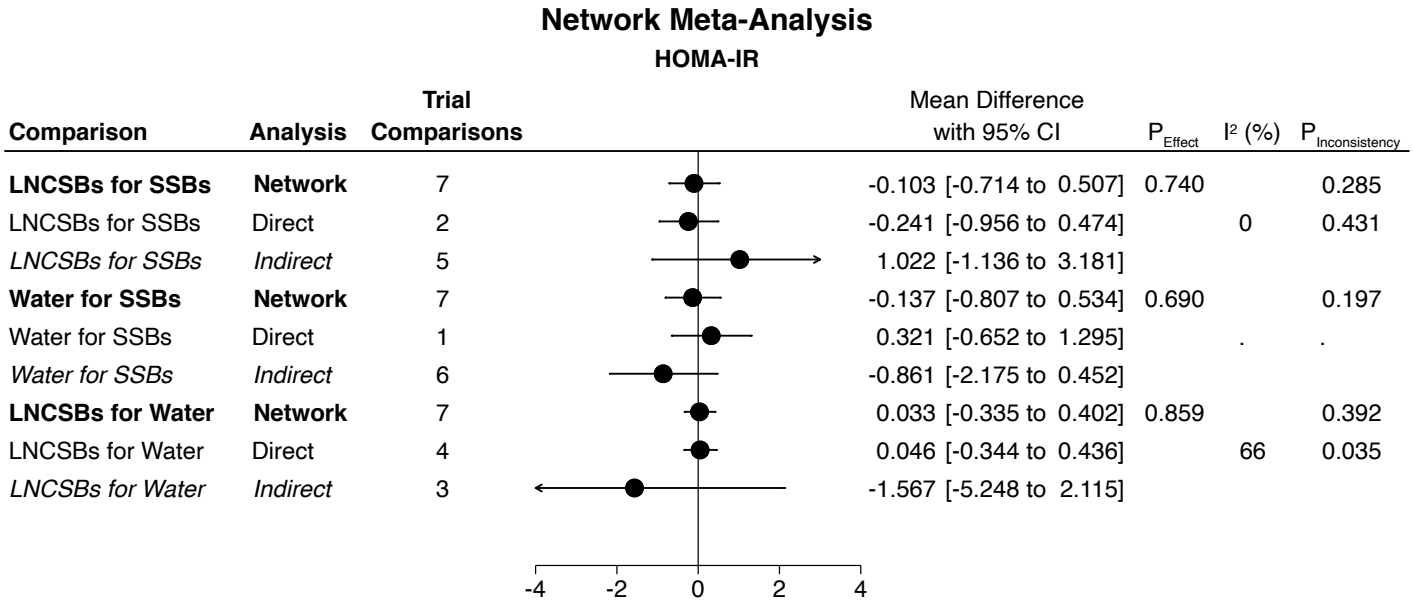
eFigure 15. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on HOMA-IR



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall
LNCSBs for SSBs	Not serious	Serious *Rated down the evidence for incoherence via visual inspection.*	Not serious	Not serious *(MID for HOMA-IR.= 1)*	NA	**MODERATE**
Water for SSBs	Not serious	Serious *Rated down the evidence for incoherence via visual inspection.*	Serious *Rated down for indirectness as only 1 direct study available.*	Not serious *(MID for HOMA-IR.= 1)*	NA	**LOW**
LNCSBs for Water	Not serious	Very Serious *Rated down for inconsistency assessed in the direct estimates as substantial unexplained heterogeneity (I2 >50%, P<0.10)* *Rated down for incoherence via visual inspection.*	Not serious	Not serious *(MID for HOMA-IR.= 1)*	NA	**LOW**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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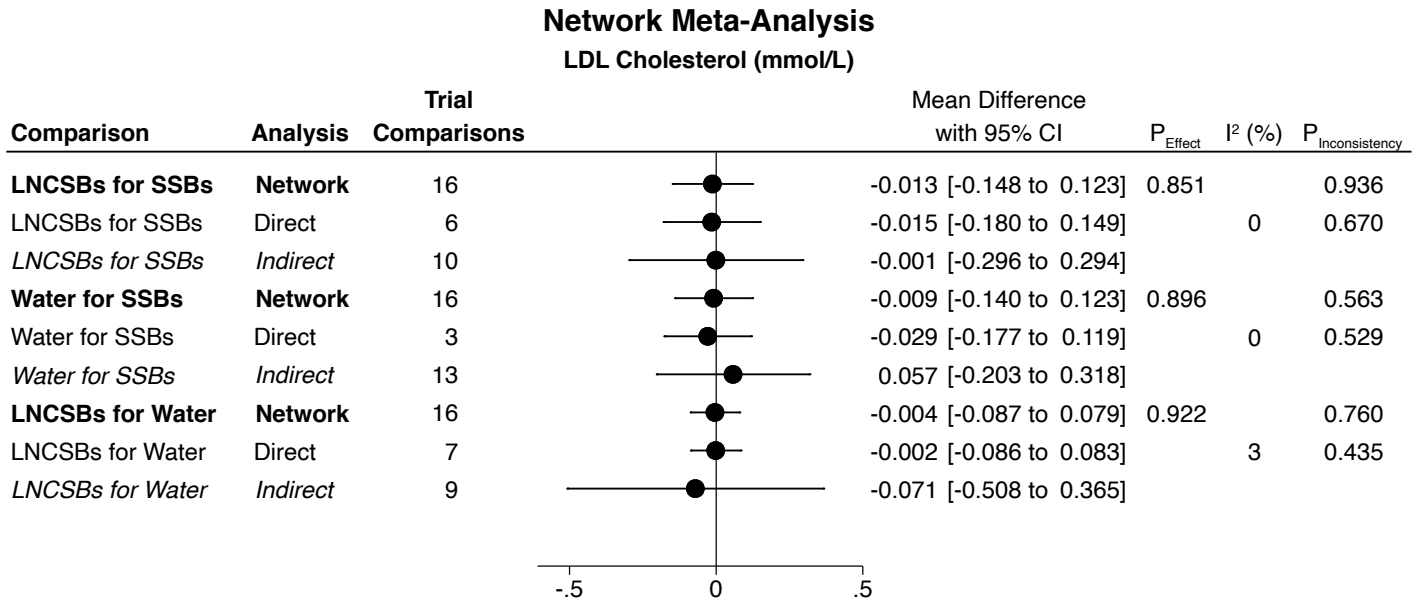
eFigure 16. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on LDL-C



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall
LNCSBs for SSBs	Not serious	Not serious	Not serious.	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for LDL-C = 0.1 mmol/L)*	Not serious	**MODERATE**
Water for SSBs	Not serious	Serious *Rated down for incoherence via visual inspection.*	Not serious.	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for LDL-C = 0.1 mmol/L)*	Not serious	**LOW**
LNCSBs for Water	Not serious	Not serious	Not serious.	Not serious *(MIDs for LDL-C = 0.1 mmol/L)*	Not serious	**HIGH**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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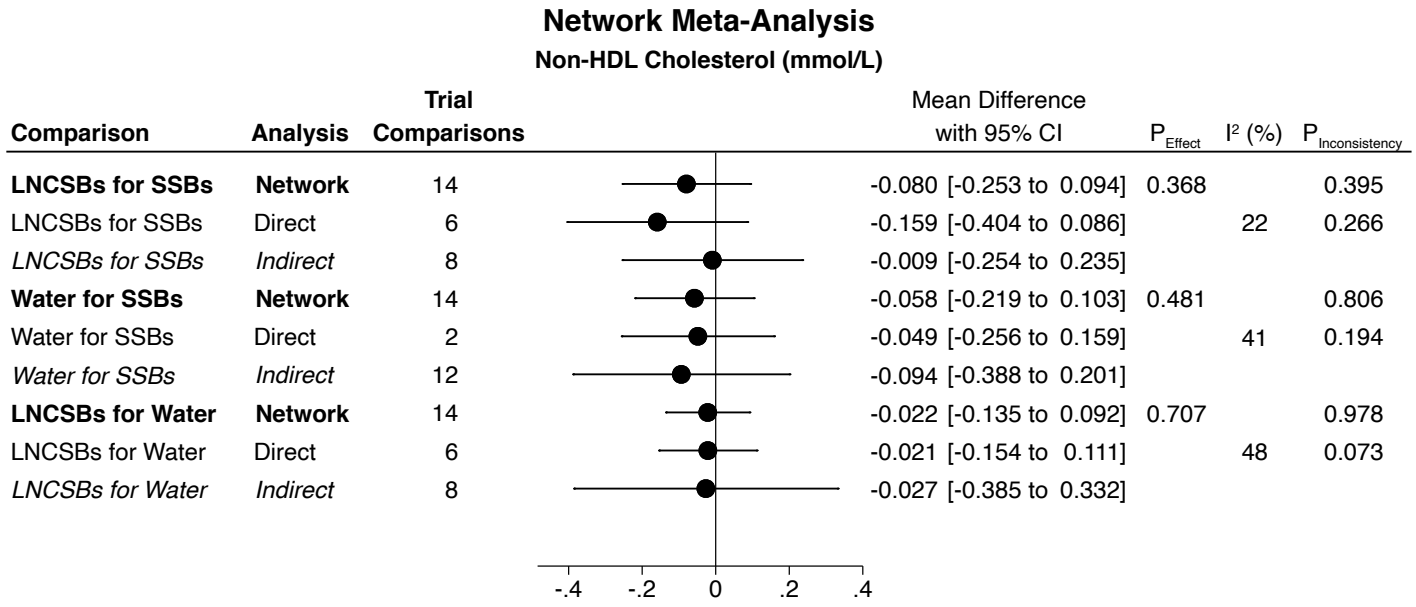
eFigure 17. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on Non-HDL-C



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall
LNCSBs for SSBs	Not serious	Not serious	Not serious.	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for Non-HDL-C = 0.1 mmol/L)*	NA	**MODERATE**
Water for SSBs	Not serious	Not serious	Not serious.	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for Non-HDL-C = 0.1 mmol/L)*	NA	**MODERATE**
LNCSBs for Water	Not serious	Not serious	Not serious.	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for Non-HDL-C = 0.1 mmol/L)*	NA	**MODERATE**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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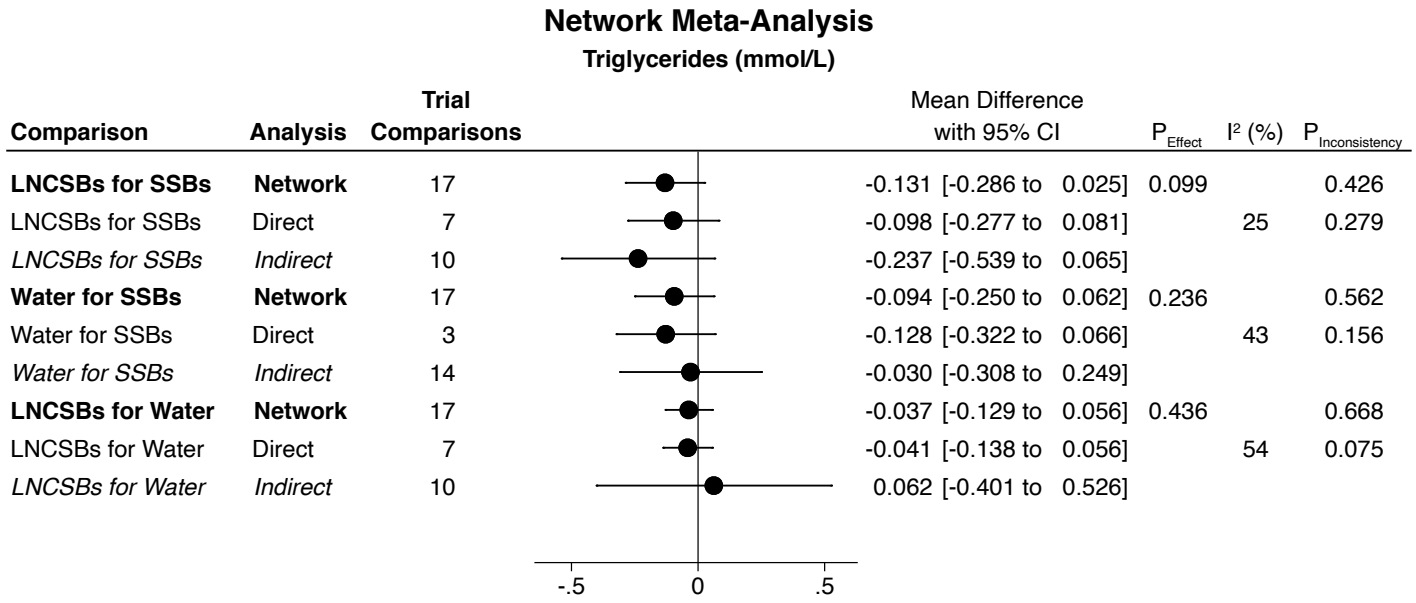
eFigure 18. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on Triglycerides



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall
LNCSBs for SSBs	Not serious	Not serious	Not serious.	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for TG = 0.1 mmol/L)*	Not serious	**MODERATE**
Water for SSBs	Not serious	Not serious	Not serious.	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for TG = 0.1 mmol/L)*	Not serious	**MODERATE**
LNCSBs for Water	Not serious	Very serious *Rated down for inconsistency assessed in the direct estimates as substantial unexplained heterogeneity (I2 >50%, P<0.10)* *Rated down for incoherence via visual inspection.*	Not serious	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for TG = 0.1 mmol/L)*	Not serious	**VERY LOW**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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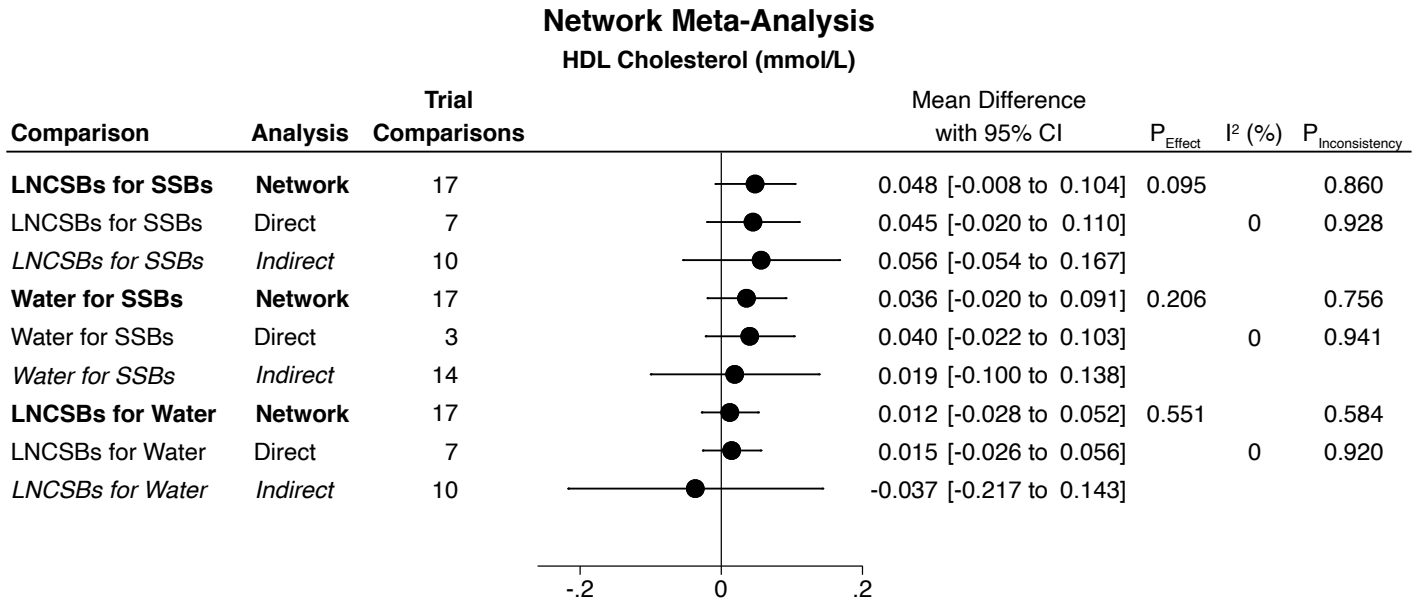
eFigure 19. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on HDL-C



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall
LNCSBs for SSBs	Not serious	Not serious	Not serious.	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for HDL-C = 0.1 mmol/L)*	Not serious	**MODERATE**
Water for SSBs	Not serious	Not serious	Not serious.	Not serious *(MIDs for HDL-C = 0.1 mmol/L)*	Not serious	**HIGH**
LNCSBs for Water	Not serious	Serious *Rated down for incoherence via visual inspection.*	Not serious	Not serious *MIDs for HDL-C = 0.1 mmol/L)*	Not serious	**LOW**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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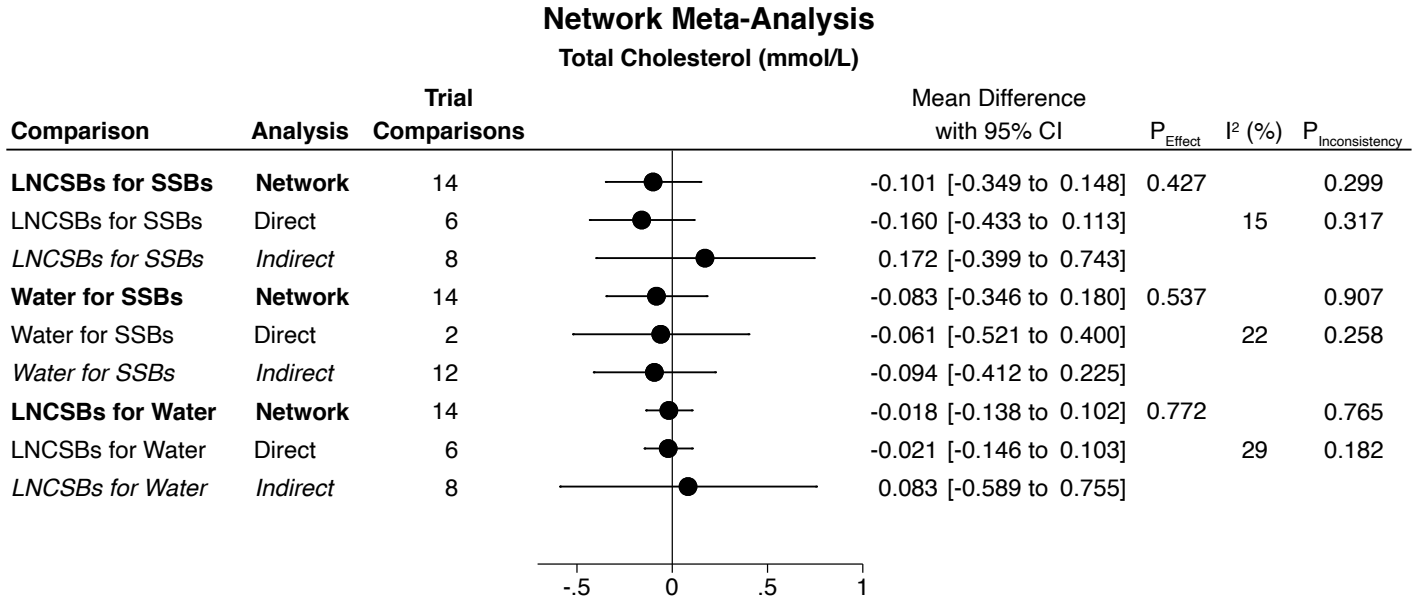
eFigure 20. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on total cholesterol



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall
LNCSBs for SSBs	Not serious	Serious *Rated down for incoherence via visual inspection.*	Not serious.	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for TC = 0.1 mmol/L)*	Not serious	**LOW**
Water for SSBs	Not serious	Not serious	Not serious.	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for TC = 0.1 mmol/L)*	Not serious	**MODERATE**
LNCSBs for Water	Not serious	Serious *Rated down for incoherence via visual inspection.*	Not serious	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for TC = 0.1 mmol/L)*	Not serious	**LOW**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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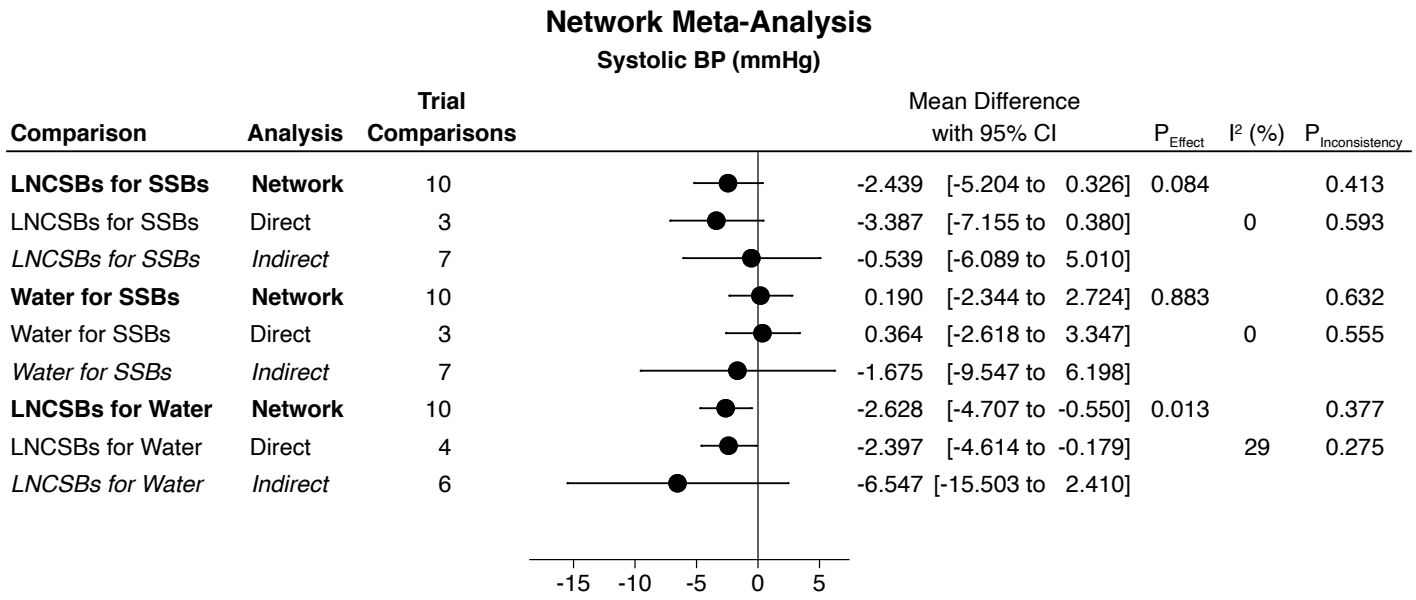
eFigure 21. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on SBP



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall
LNCSBs for SSBs	Not serious	Not serious	Not serious.	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for SBP = 2 mmHg)*	Not serious	**MODERATE**
Water for SSBs	Not serious	Serious *Rated down for incoherence via visual inspection.*	Not serious.	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for SBP = 2 mmHg)*	Not serious	**LOW**
LNCSBs for Water	Not serious	Serious *Rated down for incoherence via visual inspection*	Not serious	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for SBP = 2 mmHg)*	Not serious	**LOW**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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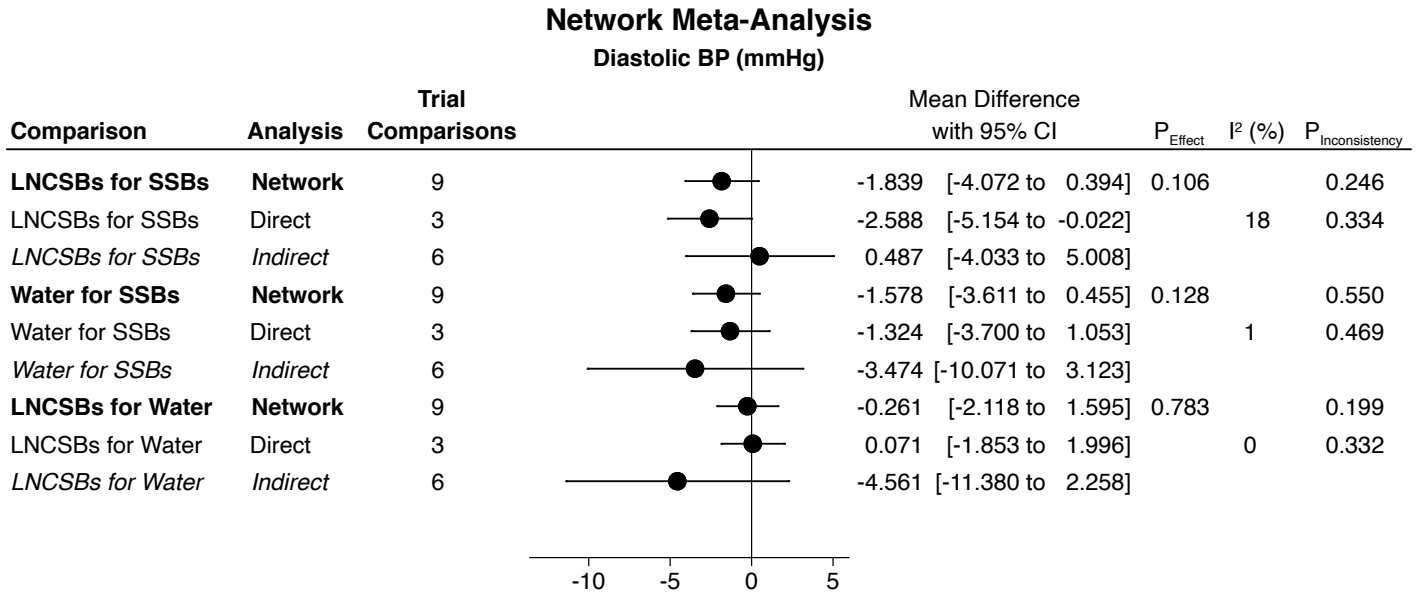
eFigure 22. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on DBP



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall
LNCSBs for SSBs	Not serious	Serious *Rated down for incoherence via visual inspection*	Not serious.	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for DBP = 2 mmHg)*	NA	**LOW**
Water for SSBs	Not serious	Not serious	Not serious.	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for DBP = 2 mmHg)*	NA	**MODERATE**
LNCSBs for Water	Not serious	Serious *Rated down for incoherence via visual inspection*	Not serious	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for DBP = 2 mmHg)*	**NA**	**LOW**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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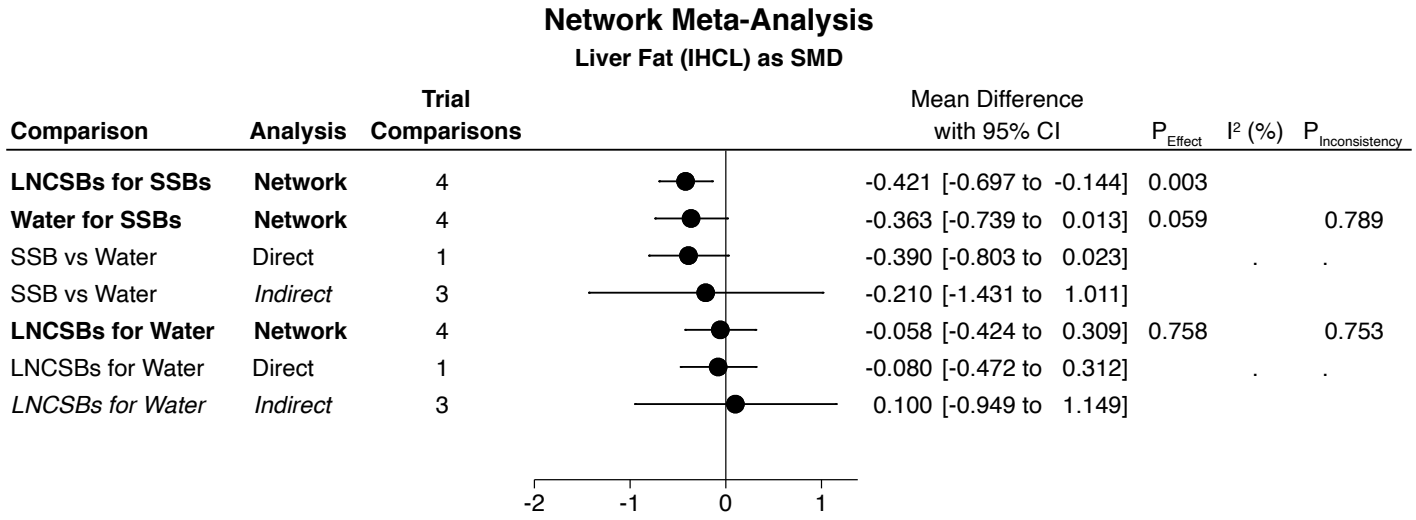
eFigure 23. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on IHCL



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall
LNCSBs for SSBs	Not serious	Not serious	Not serious	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for IHCL = 0.1)*	NA	**MODERATE**
Water for SSBs	Not serious	Not serious	Serious *Rated down for indirectness as only 1 direct study available.*	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for IHCL = 0.1)*	NA	**LOW**
LNCSBs for Water	Not serious	Serious *Rated down for incoherence via visual inspection*	Serious *Rated down for indirectness as only 1 direct study available.*	Not serious	NA	**LOW**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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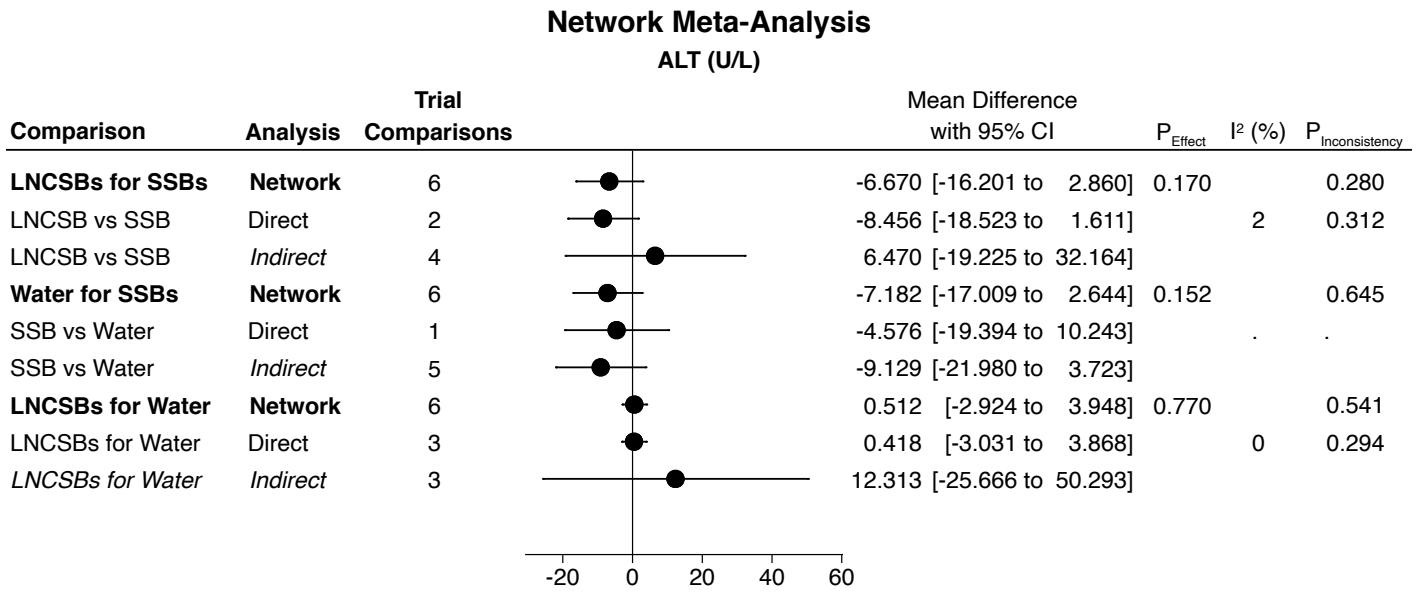
eFigure 24. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on ALT



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall
LNCSBs for SSBs	Not serious	Serious *Rated down for incoherence via visual inspection*	Not serious.	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for ALT = 1)*	NA	**LOW**
Water for SSBs	Not serious	Not serious	Serious *Rated down for indirectness as only 1 direct study available.*	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for ALT = 1)*	NA	**LOW**
LNCSBs for Water	Not serious	Not serious	Not serious	Not serious	NA	**LOW**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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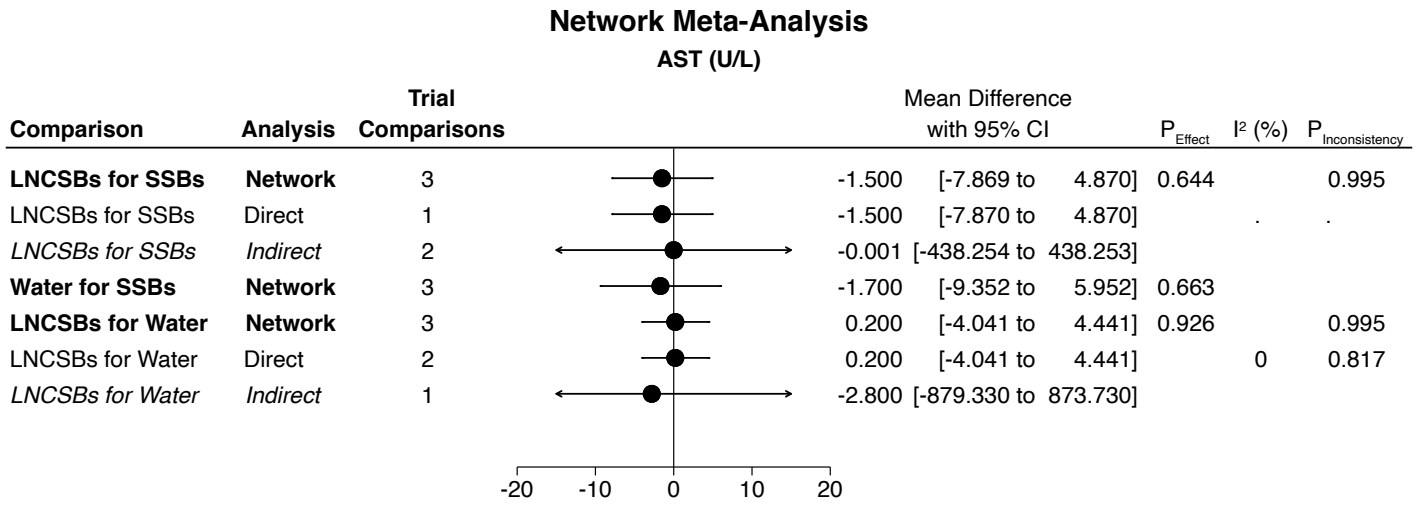
eFigure 25. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on AST



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall
LNCSBs for SSBs	Not serious	Not serious	Serious *Rated down for indirectness as only 1 direct study available.*	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for AST = 1)*	NA	**LOW**
Water for SSBs	Not serious	Not serious	Serious *Rated down for indirectness as evidence only comes from indirect studies..*	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for AST = 1)*	NA	**LOW**
LNCSBs for Water	Not serious	Serious *Rated down for incoherence via visual inspection*	Not serious	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for AST = 1)*	NA	**LOW**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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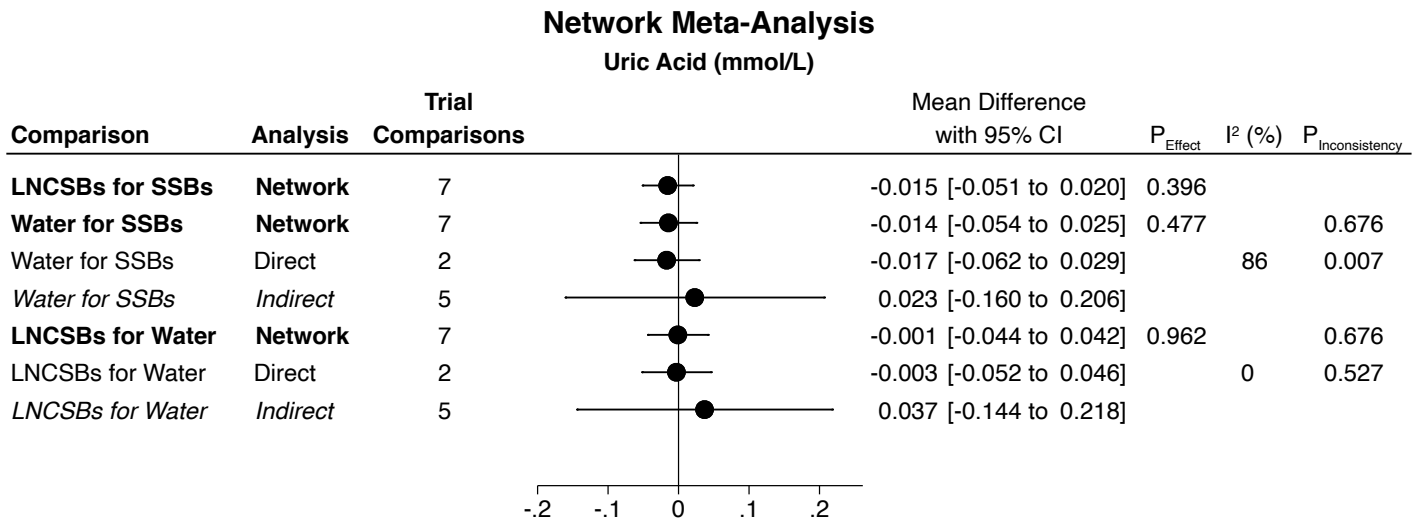
eFigure 26. Network analysis with GRADE assessment of the certainty of the evidence comparing LNCSBs, SSBs and Water on uric acid



Using restricted maximum likelihood (REML) random-effects model

Note: $P_{Inconsistency}$ represents P-value for *incoherence factor* in the network estimates, and for *heterogeneity* in the direct pair-wise estimates

GRADE	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall
LNCSBs for SSBs	Not serious	Not serious	Not serious	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for Uric acid = 0.013 mmol/L)*	NA	**MODERATE**
Water for SSBs	Not serious	Very Serious *Rated down for inconsistency assessed in the direct estimates as substantial unexplained heterogeneity (I2 >50%, P<0.10)* *Rated down for incoherence via visual inspection*	Not serious	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for Uric acid = 0.013 mmol/L)*	NA	**VERY LOW**
LNCSBs for Water	Not serious	Very Serious *Rated down for inconsistency assessed in the direct estimates as substantial unexplained heterogeneity (I2 >50%, P<0.10)* *Rated down for incoherence via visual inspection*	Not serious	Serious *Rated down for imprecision when the 95% CI were wide and crossed prespecified MIDs for benefit and/or harm. (MIDs for Uric acid = 0.013 mmol/L)*	NA	**VERY LOW**

GRADE, Grading of Recommendations, Assessment, Development, and Evaluation; MID, Minimally Important Difference; NA, not available; LNCSBs, low- and no-calorie sweetened beverages; SSBs, sugar-sweetened beverages.

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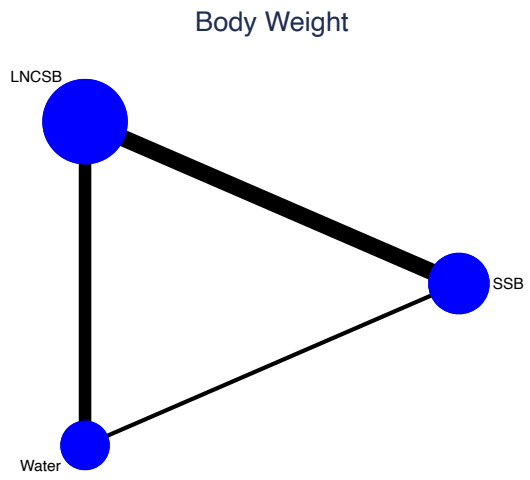
eFigure 27. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with Body Weight



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages.

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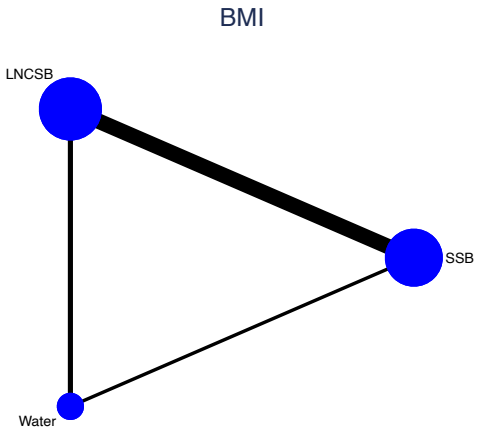
eFigure 28. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with BMI



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. BMI; body mass index, LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages.

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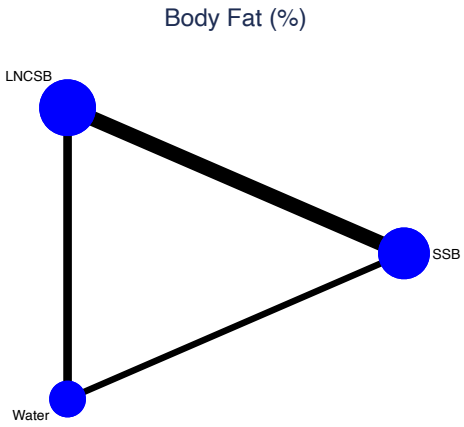
eFigure 29. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with Body Fat %



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages.

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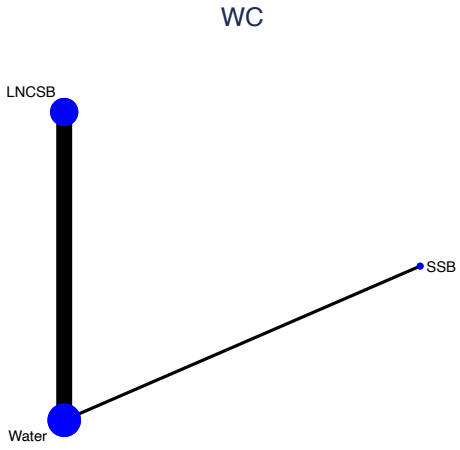
eFigure 30. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with Waist Circumference



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages.

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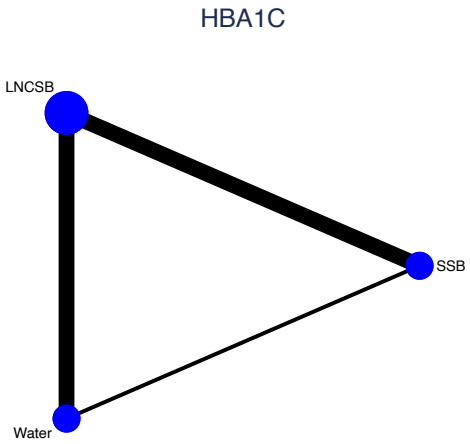
eFigure 31. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with HbA1c



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. HbA1c, hemoglobin A1c; LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages.

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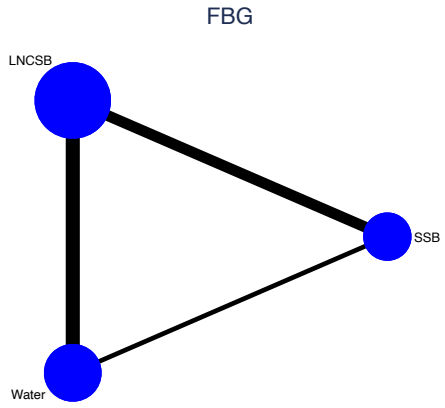
eFigure 32. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with FPG



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. FPG, fasting plasma glucose; LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages.

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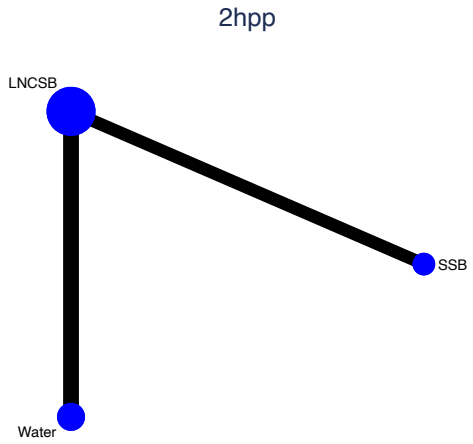
eFigure 33. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with 2h-PP



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. 2h-PP, two-hour post prandial glucose; LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages.

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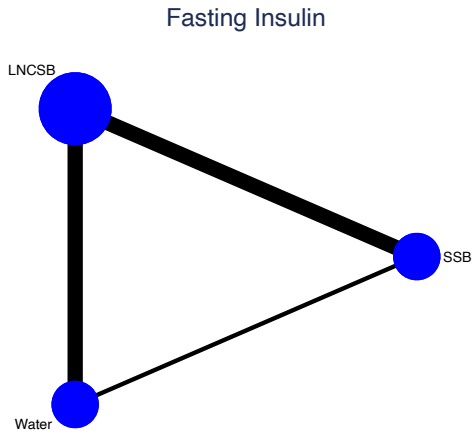
eFigure 34. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with FPI



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. 2h-PP, two-hour post prandial glucose; LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages.

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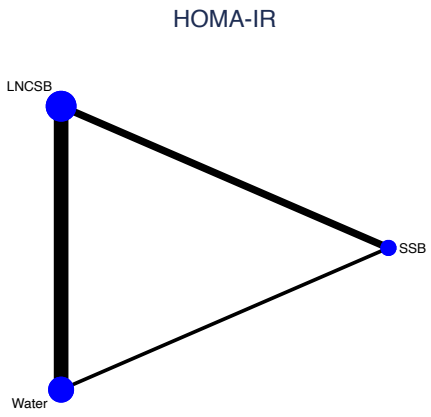
eFigure 35. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with HOMA-IR



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. HOMA-IR; homeostatic model assessment of insulin resistance; LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages.

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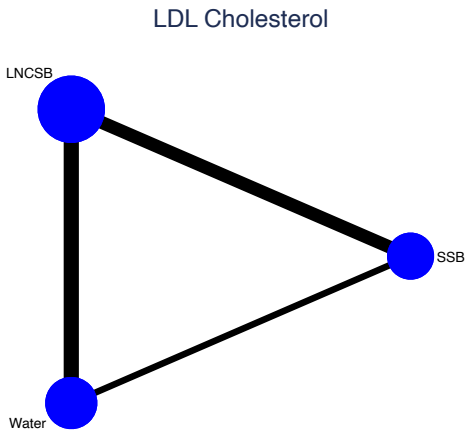
eFigure 36. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with LDL-C



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. LDL-C, low-density lipoprotein cholesterol; LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages.

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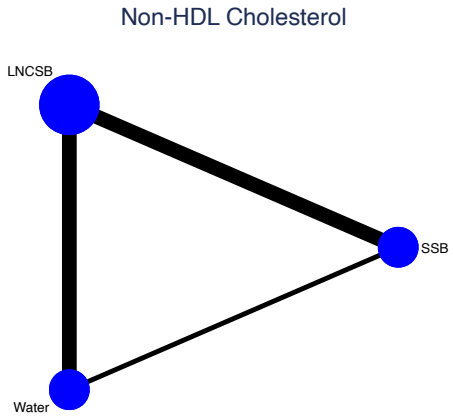
eFigure 37. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with Non-HDL-C



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. Non-HDL-C, non-high-density lipoprotein cholesterol, LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages.

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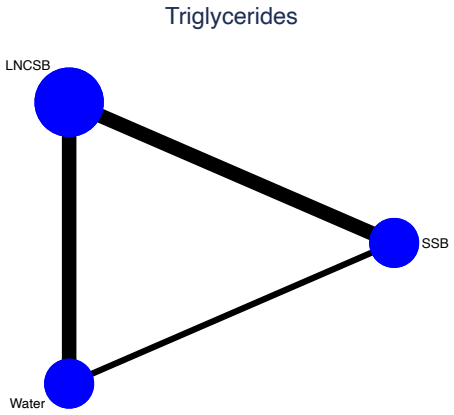
eFigure 38. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with TGs



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages; TGs, triglycerides.

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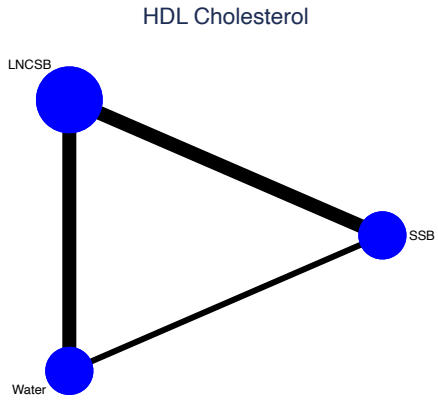
eFigure 39. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with HDL-C



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. HDL-C, high-density lipoprotein cholesterol, LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages.

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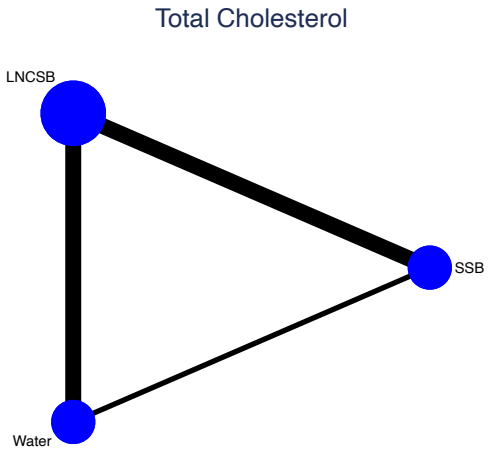
eFigure 40. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with TC



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages; TC, total cholesterol.

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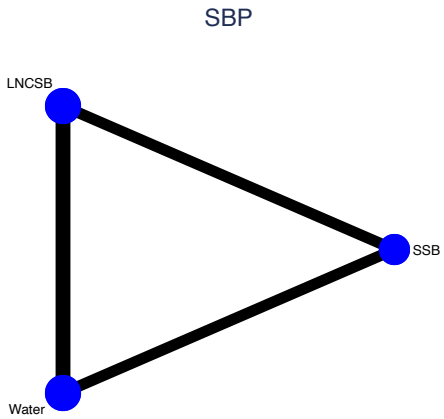
eFigure 41. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with SBP



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. LNCSBs, non-nutritive sweetened beverages; SBP, systolic blood pressure; SSBs, sugar-sweetened beverages.

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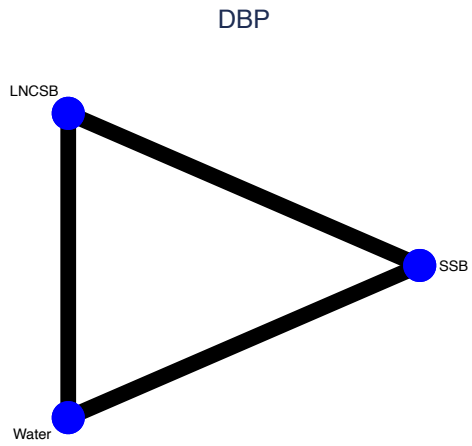
eFigure 42. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with DBP



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. DBP, diastolic blood pressure; LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages.

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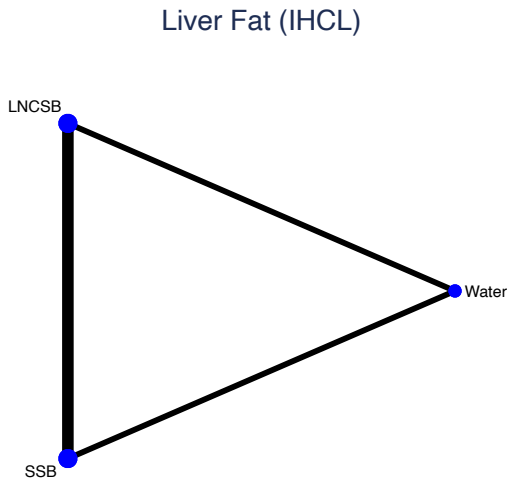
eFigure 43. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with IHCL



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. IHCL, intra-hepatocellular lipid; LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages.

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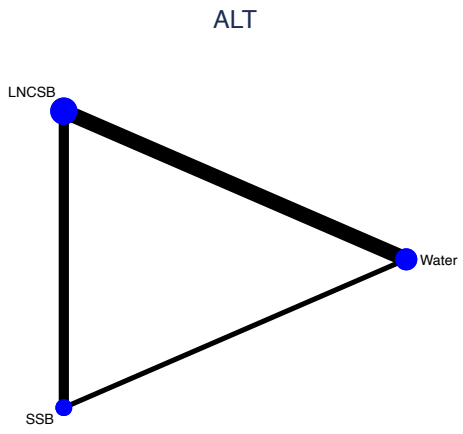
eFigure 44. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with ALT



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. ALT, alanine transaminase; LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages.

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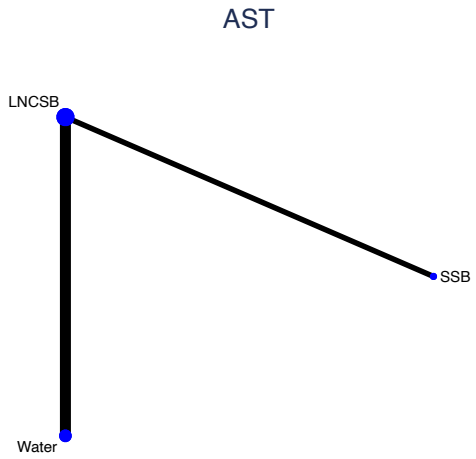
eFigure 45. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with AST



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. AST, aspartate transaminase; LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages.

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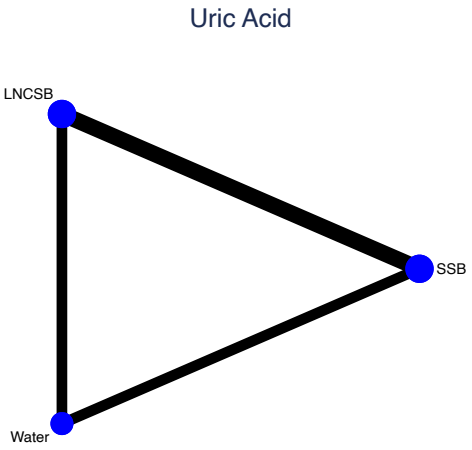
eFigure 46. Network diagram for randomized controlled trials investigating the association of the substitution of LNCSBs for SSBs, water for SSBs, and LNCSBs for water with uric acid



Blue nodes represent the study size for each beverage under investigation. The thickness of the black lines represents the number of studies directly comparing one beverage to another. LNCSBs, non-nutritive sweetened beverages; SSBs, sugar-sweetened beverages.

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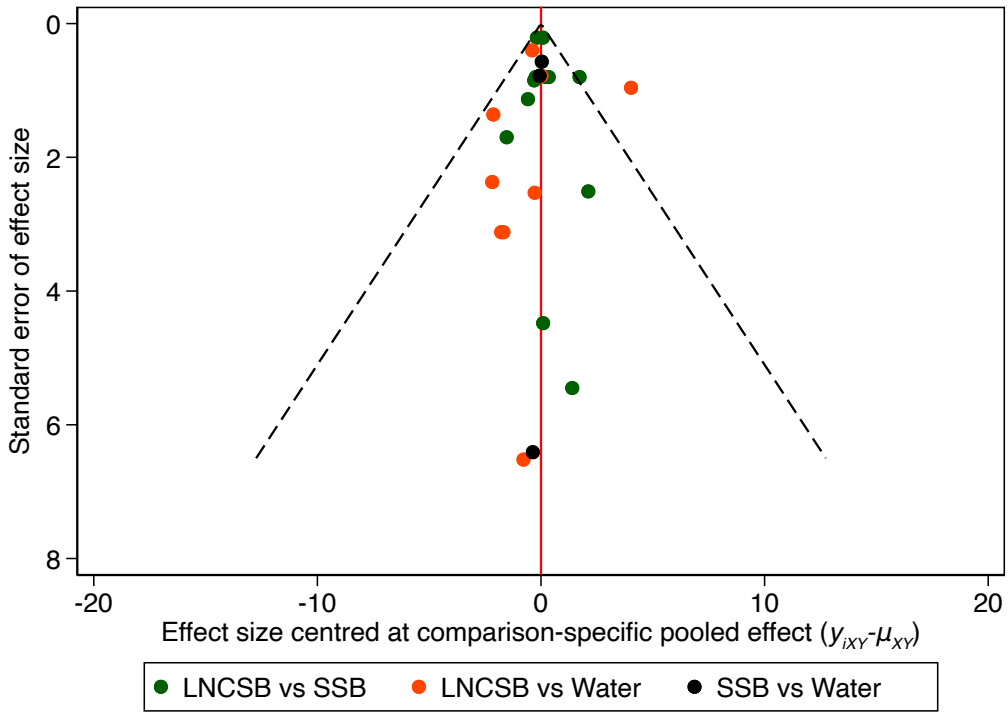
Publication Bias

eFigure 47. Comparison adjusted funnel plot for Body Weight



The horizontal axis represents an adjusted effect size, presenting the difference between each observed effect size and the mean effect size for the specific comparison being made. The dashed lines represent pseudo 95% confidence intervals.

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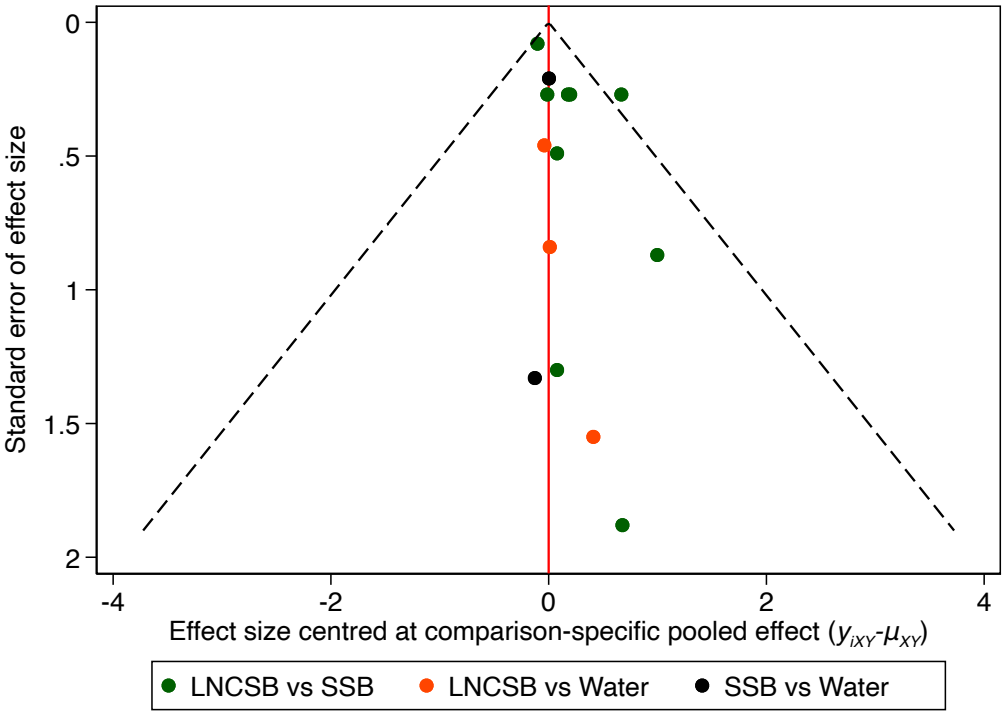
eFigure 48. Comparison adjusted funnel plot for BMI



The horizontal axis represents an adjusted effect size, presenting the difference between each observed effect size and the mean effect size for the specific comparison being made. The dashed lines represent pseudo 95% confidence intervals.

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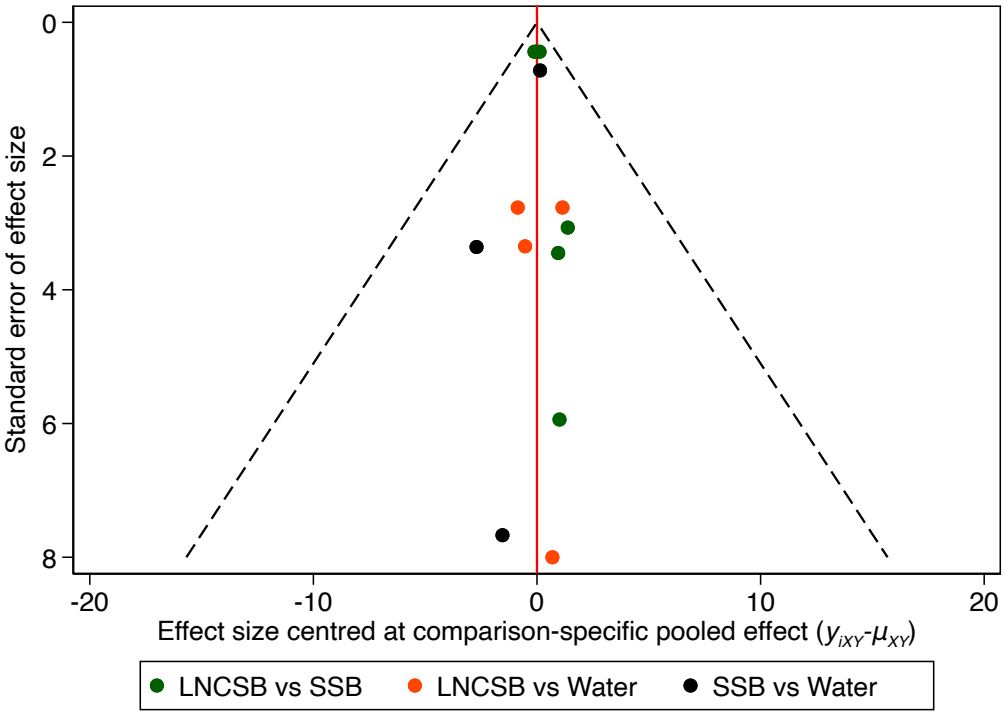
eFigure 49. Comparison adjusted funnel plot for Body Fat (%)



The horizontal axis represents an adjusted effect size, presenting the difference between each observed effect size and the mean effect size for the specific comparison being made. The dashed lines represent pseudo 95% confidence intervals.

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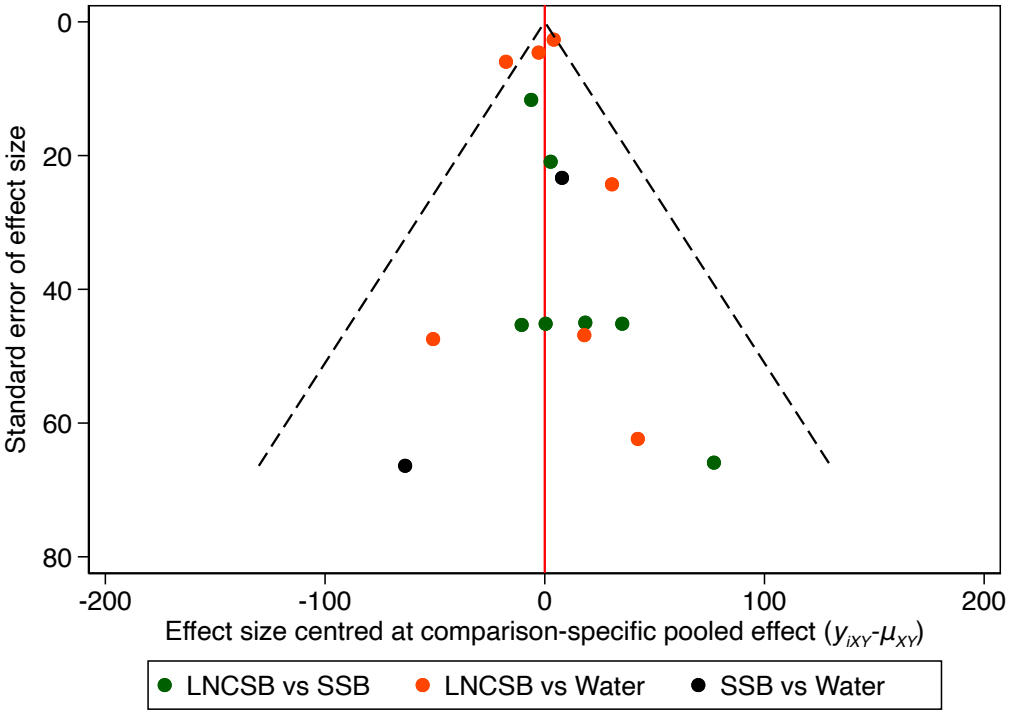
eFigure 50. Comparison adjusted funnel plot for Fasting Insulin



The horizontal axis represents an adjusted effect size, presenting the difference between each observed effect size and the mean effect size for the specific comparison being made. The dashed lines represent pseudo 95% confidence intervals.

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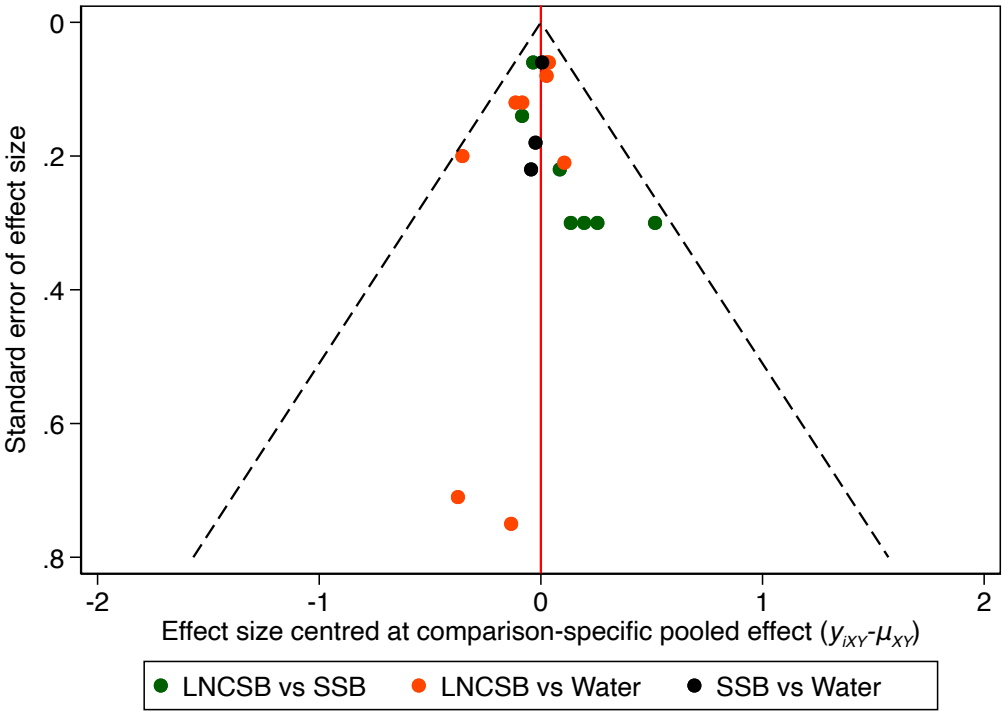
eFigure 51. Comparison adjusted funnel plot for Fasting Blood Glucose



The horizontal axis represents an adjusted effect size, presenting the difference between each observed effect size and the mean effect size for the specific comparison being made. The dashed lines represent pseudo 95% confidence intervals.

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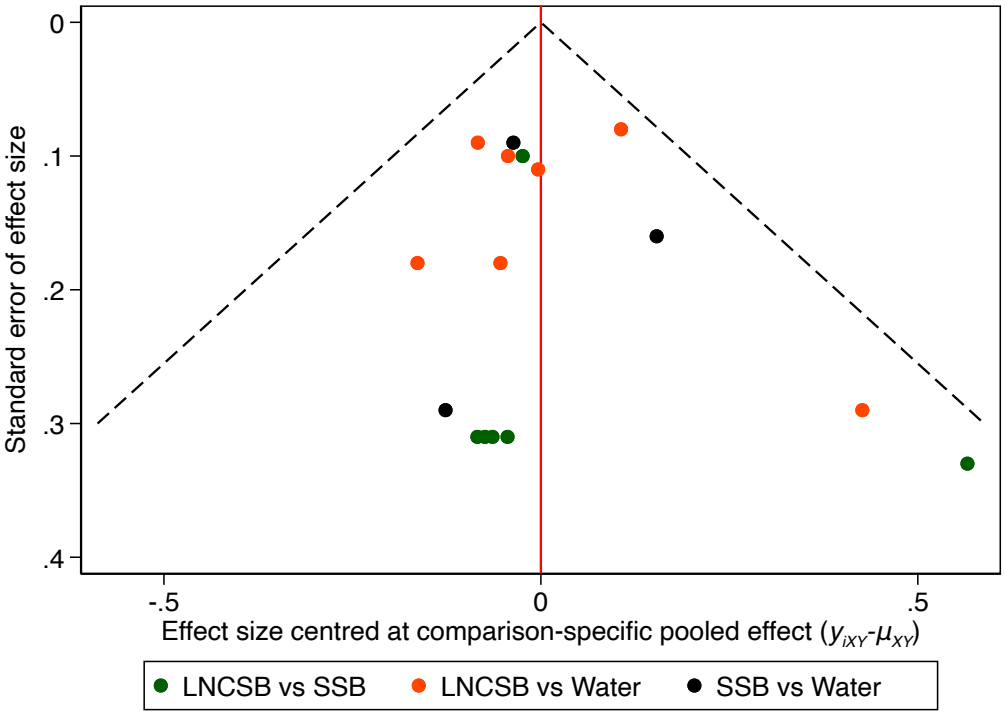
eFigure 52. Comparison adjusted funnel plot for LDL-Cholesterol



The horizontal axis represents an adjusted effect size, presenting the difference between each observed effect size and the mean effect size for the specific comparison being made. The dashed lines represent pseudo 95% confidence intervals.

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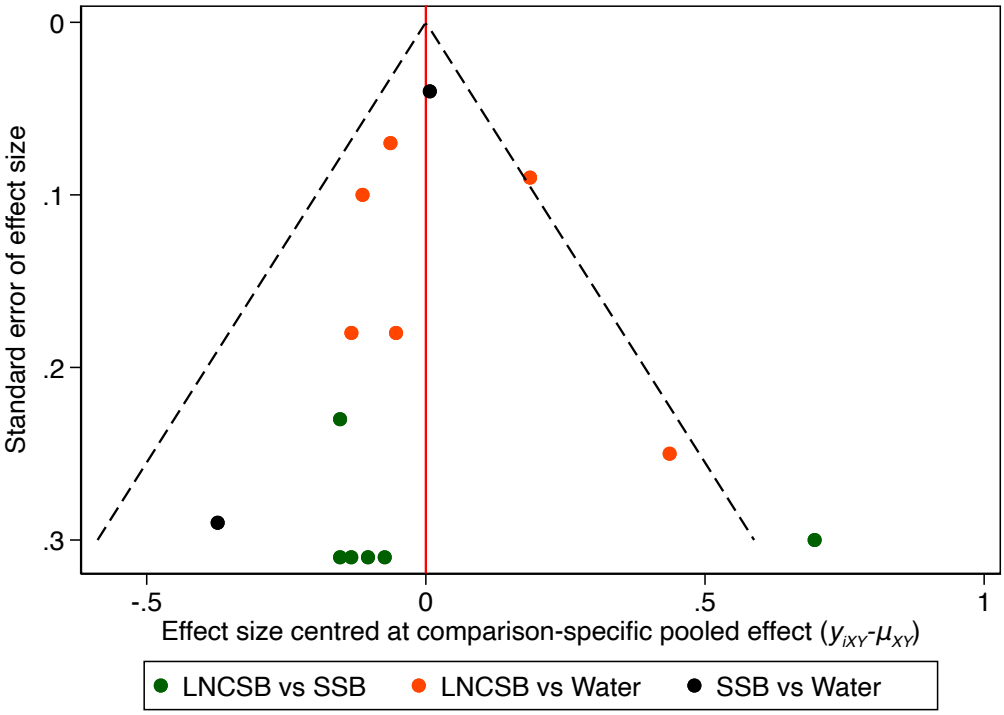
eFigure 53. Comparison adjusted funnel plot for Non-HDL-Cholesterol



The horizontal axis represents an adjusted effect size, presenting the difference between each observed effect size and the mean effect size for the specific comparison being made. The dashed lines represent pseudo 95% confidence intervals.

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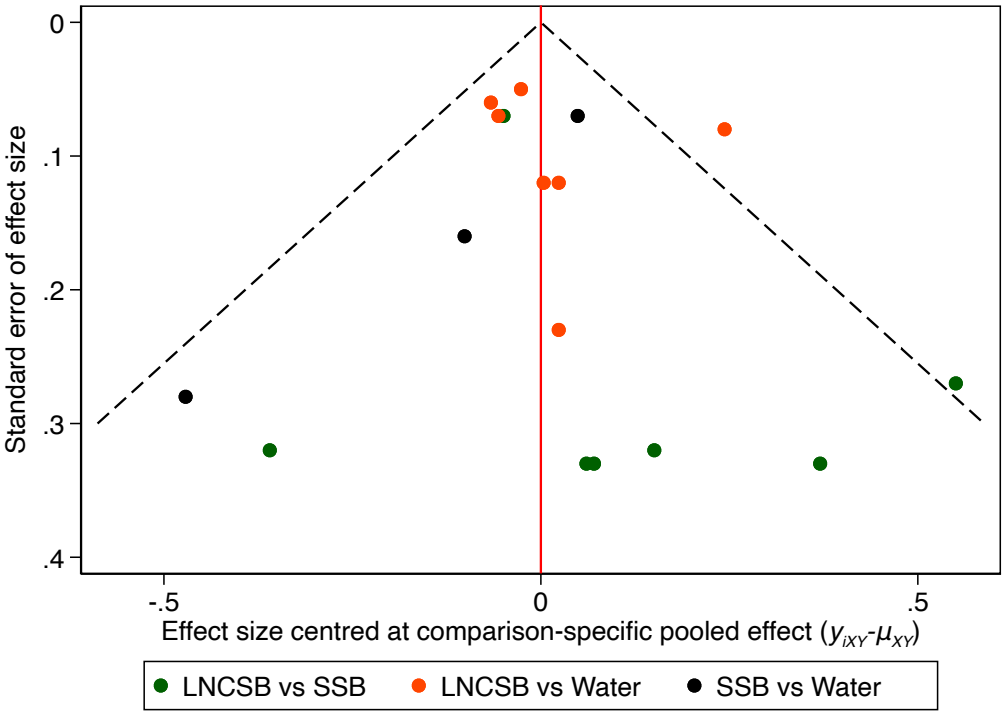
eFigure 54. Comparison adjusted funnel plot for Triglycerides



The horizontal axis represents an adjusted effect size, presenting the difference between each observed effect size and the mean effect size for the specific comparison being made. The dashed lines represent pseudo 95% confidence intervals.

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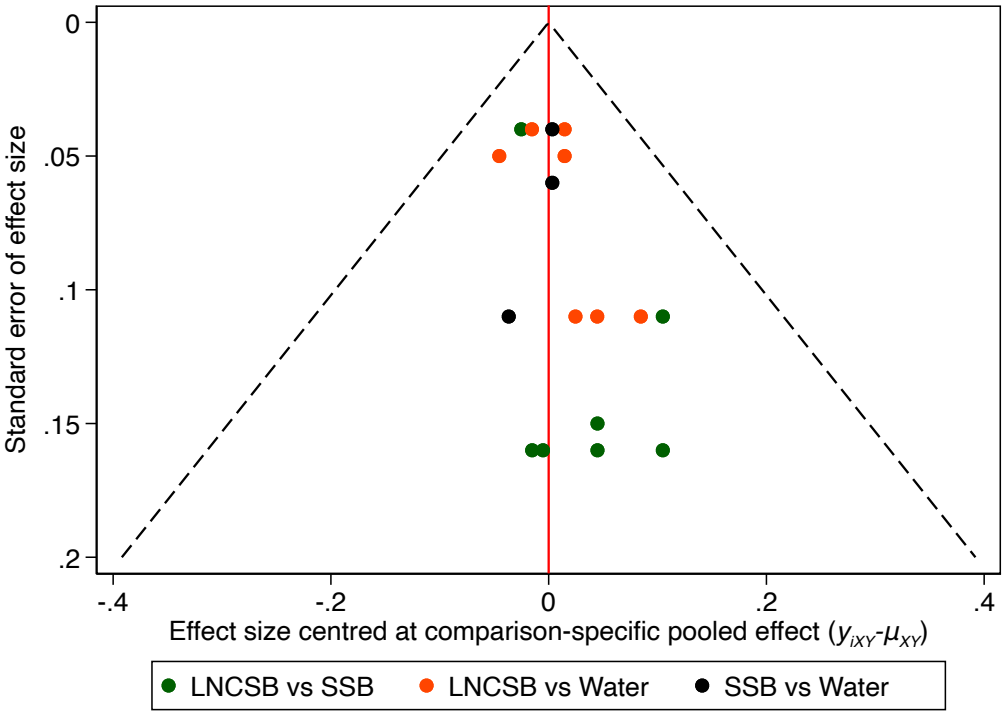
eFigure 55. Comparison adjusted funnel plot for HDL-Cholesterol



The horizontal axis represents an adjusted effect size, presenting the difference between each observed effect size and the mean effect size for the specific comparison being made. The dashed lines represent pseudo 95% confidence intervals.

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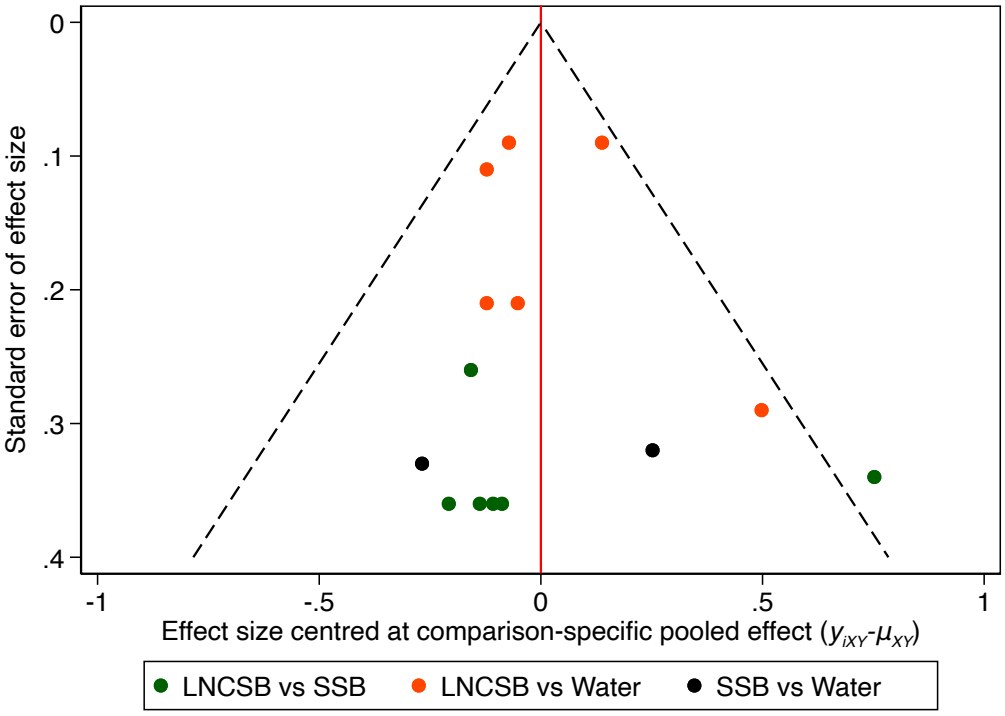
eFigure 56. Comparison adjusted funnel plot for Total-Cholesterol



The horizontal axis represents an adjusted effect size, presenting the difference between each observed effect size and the mean effect size for the specific comparison being made. The dashed lines represent pseudo 95% confidence intervals.

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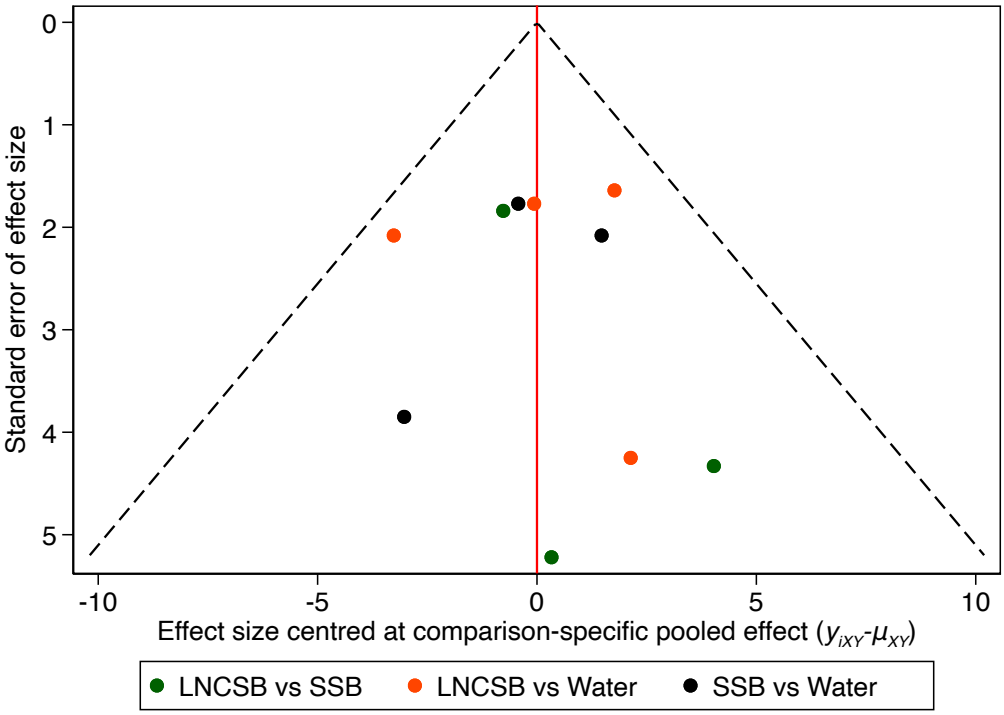
eFigure 57. Comparison adjusted funnel plot for Systolic Blood Pressure



The horizontal axis represents an adjusted effect size, presenting the difference between each observed effect size and the mean effect size for the specific comparison being made. The dashed lines represent pseudo 95% confidence intervals.

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Mean Differences and Standard Errors

eData 1. Body Weight (Kg)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Bonnet 2018	Water	LNCSB	-0.59	0.4
Campos 2015	SSB	LNCSB	-2.3	5.45
Ebbeling 2020	Water	LNCSB	-1	0.78
Ebbeling 2020	SSB	LNCSB	-0.6	0.85
Ebbeling 2020	SSB	Water	-0.5	0.78
Engel 2018	Water	LNCSB	-0.2	6.52
Engel 2018	SSB	LNCSB	-1	4.48
Engel 2018	SSB	Water	-0.80	6.41
Hernandez-Cordero 2014	SSB	Water	-0.4	0.57
Higgins 2018 (1050mg ASP)	Water	LNCSB	0.7	3.12
Higgins 2018 (350mg ASP)	Water	LNCSB	0.8	3.12
Higgins 2019 (Asp)	SSB	LNCSB	-1.12	0.8
Higgins 2019 (RebA)	SSB	LNCSB	-1.25	0.8
Higgins 2019 (Sacrn)	SSB	LNCSB	-0.67	0.8
Higgins 2019 (Sucralose)	SSB	LNCSB	-2.63	0.8
Madjd 2015	Water	LNCSB	1.20	2.37
Madjd 2017	Water	LNCSB	1.15	1.36
Peters 2016	Water	LNCSB	-5.01	0.96
Reid 2007	SSB	LNCSB	0.63	1.7
Reid 2010	SSB	LNCSB	-0.32	1.13
Reid 2014	SSB	LNCSB	-3.02	2.51
Tate 2012	Water	LNCSB	-0.70	2.53
Tordoff 1990 (Females)	SSB	LNCSB	-0.72	0.21
Tordoff 1990 (Males)	SSB	LNCSB	-0.99	0.21

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eData 2. BMI (kg/m^2)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Campos 2015	SSB	LNCSB	-0.9	1.88
Engel 2018	Water	LNCSB	0	1.55
Engel 2018	SSB	LNCSB	-0.3	1.30
Engel 2018	SSB	Water	-0.3	1.33
Hernandez-Cordero 2014	SSB	Water	-0.17	0.21
Higgins 2019 (Asp)	SSB	LNCSB	-0.4	0.27
Higgins 2019 (RebA)	SSB	LNCSB	-0.42	0.27
Madjd 2015	Water	LNCSB	0.40	0.84
Madjd 2017	Water	LNCSB	0.45	0.46
Reid 2007	SSB	LNCSB	-0.3	0.49
Reid 2010	SSB	LNCSB	-0.12	0.08
Reid 2014	SSB	LNCSB	-1.22	0.87

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eData 3. Body Fat (%)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Campos 2015	SSB	LNCSB	-2	3.07
Ebbeling 2020	Water	LNCSB	1.27	3.35
Ebbeling 2020	SSB	LNCSB	-1.57	3.45
Ebbeling 2020	SSB	Water	-2.84	3.36
Engel 2018	Water	LNCSB	0.05	8.00
Engel 2018	SSB	LNCSB	-1.63	5.94
Engel 2018	SSB	Water	-1.68	7.67
Hernandez-Cordero 2014	SSB	Water	0	0.72
Higgins 2018 (1050mg ASP)	Water	LNCSB	-0.4	2.77
Higgins 2018 (350mg ASP)	Water	LNCSB	1.6	2.77
Higgins 2019 (Asp)	SSB	LNCSB	-0.51	0.44
Higgins 2019 (RebA)	SSB	LNCSB	-0.61	0.44
Higgins 2019 (Sacrn)	SSB	LNCSB	-0.58	0.44
Higgins 2019 (Sucralose)	SSB	LNCSB	-0.74	0.44

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eData 4. Waist Circumference (cm)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Bonnet 2018	Water	LNCSB	0.2	0.66
Hernandez-Cordero 2014	SSB	Water	0.3	0.81
Madjd 2015	Water	LNCSB	0.7	1.56
Madjd 2017	Water	LNCSB	0	1.42
Peters 2016	Water	LNCSB	-4.5	1.16
Tate 2012	Water	LNCSB	-0.1	2.01

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eData 5. HbA1c (%)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Hernandez-Cordero 2014	SSB	Water	-0.05	0.04
Higgins 2018 (1050mg ASP)	Water	LNCSB	0.01	0.07
Higgins 2018 (350mg ASP)	Water	LNCSB	0.03	0.07
Higgins 2019 (Asp)	SSB	LNCSB	0.09	0.13
Higgins 2019 (RebA)	SSB	LNCSB	0.12	0.13
Higgins 2019 (Sacrn)	SSB	LNCSB	0.14	0.13
Madjd 2015	Water	LNCSB	0.20	0.11
Madjd 2017	Water	LNCSB	0.75	0.13

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eData 6. Fasting Blood Glucose (mmol/L)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Bonnet 2018	Water	LNCSB	-0.04	0.06
Campos 2015	SSB	LNCSB	0.00	0.14
Ebbeling 2020	Water	LNCSB	-0.05	0.06
Ebbeling 2020	SSB	LNCSB	-0.05	0.06
Ebbeling 2020	SSB	Water	0	0.06
Engel 2018	Water	LNCSB	-0.12	0.21
Engel 2018	SSB	LNCSB	-0.17	0.22
Engel 2018	SSB	Water	-0.05	0.22
Hernandez-Cordero 2014	SSB	Water	-0.03	0.18
Higgins 2018 (1050mg ASP)	Water	LNCSB	0.12	0.75
Higgins 2018 (350mg ASP)	Water	LNCSB	0.36	0.71
Higgins 2019 (Asp)	SSB	LNCSB	-0.34	0.3
Higgins 2019 (RebA)	SSB	LNCSB	-0.22	0.3
Higgins 2019 (Sacrn)	SSB	LNCSB	-0.60	0.3
Higgins 2019 (Sucralose)	SSB	LNCSB	-0.28	0.3
Madjd 2015	Water	LNCSB	0.10	0.12
Madjd 2017	Water	LNCSB	0.34	0.2
Peters 2016	Water	LNCSB	-0.04	0.08
Tate 2012	Water	LNCSB	0.07	0.12

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eData 7. 2-Hour Post-prandial Glucose (mmol/L)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Bonnet 2018	Water	LNCSB	0.12	0.16
Higgins 2018 (1050mg ASP)	Water	LNCSB	-0.18	0.61
Higgins 2018 (350mg ASP)	Water	LNCSB	-0.64	0.56
Higgins 2019 (Asp)	SSB	LNCSB	0.1	0.75
Higgins 2019 (RebA)	SSB	LNCSB	0.39	0.76
Higgins 2019 (Sacrn)	SSB	LNCSB	0.26	0.76
Higgins 2019 (Sucralose)	SSB	LNCSB	0.42	0.76
Madjd 2015	Water	LNCSB	0.30	0.16
Madjd 2017	Water	LNCSB	0.31	0.24

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eData 8. Fasting Plasma Insulin (pmol/L)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Bonnet 2018	Water	LNCSB	0.78	2.66
Campos 2015	SSB	LNCSB	-15.28	20.92
Ebbeling 2020	Water	LNCSB	-37.5	62.37
Ebbeling 2020	SSB	LNCSB	-89.59	65.92
Ebbeling 2020	SSB	Water	-52.09	66.39
Engel 2018	Water	LNCSB	-25.72	24.30
Engel 2018	SSB	LNCSB	-6.4	11.67
Engel 2018	SSB	Water	19.32	23.33
Higgins 2018 (1050mg ASP)	Water	LNCSB	-13.13	46.85
Higgins 2018 (350mg ASP)	Water	LNCSB	55.69	47.44
Higgins 2019 (Asp)	SSB	LNCSB	-31.04	44.98
Higgins 2019 (RebA)	SSB	LNCSB	-12.99	45.15
Higgins 2019 (Sacrn)	SSB	LNCSB	-47.85	45.15
Higgins 2019 (Sucralose)	SSB	LNCSB	-2.08	45.32
Madjd 2015	Water	LNCSB	7.64	4.6
Madjd 2017	Water	LNCSB	22.50	5.97

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eData 9. HOMA-IR

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Bonnet 2018	Water	LNCSB	-0.19	0.10
Campos 2015	SSB	LNCSB	-0.80	0.82
Engel 2018	Water	LNCSB	-0.44	0.44
Engel 2018	SSB	LNCSB	-0.13	0.23
Engel 2018	SSB	Water	0.31	0.42
Madjd 2015	Water	LNCSB	0.20	0.18
Madjd 2017	Water	LNCSB	0.71	0.38

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eData 10. LDL-Cholesterol (mmol/L)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Ebbeling 2020	Water	LNCSB	0.08	0.09
Ebbeling 2020	SSB	LNCSB	0.01	0.1
Ebbeling 2020	SSB	Water	-0.06	0.09
Engel 2018	Water	LNCSB	-0.43	0.29
Engel 2018	SSB	LNCSB	-0.58	0.33
Engel 2018	SSB	Water	-0.15	0.29
Hernandez-Cordero 2014	SSB	Water	0.13	0.16
Higgins 2018 (1050mg ASP)	Water	LNCSB	0.16	0.18
Higgins 2018 (350mg ASP)	Water	LNCSB	0.05	0.18
Higgins 2019 (Asp)	SSB	LNCSB	0.05	0.31
Higgins 2019 (RebA)	SSB	LNCSB	0.06	0.31
Higgins 2019 (Sacrn)	SSB	LNCSB	0.07	0.31
Higgins 2019 (Sucralose)	SSB	LNCSB	0.03	0.31
Madjd 2015	Water	LNCSB	0.00	0.11
Madjd 2017	Water	LNCSB	0.04	0.1
Peters 2016	Water	LNCSB	-0.11	0.08

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eData 11. Non-HDL-Cholesterol (mmol/L)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Campos 2015	SSB	LNCSB	0.00	0.23
Engel 2018	Water	LNCSB	-0.45	0.25
Engel 2018	SSB	LNCSB	-0.85	0.3
Engel 2018	SSB	Water	-0.40	0.29
Hernandez-Cordero 2014	SSB	Water	-0.02	0.04
Higgins 2018 (1050mg ASP)	Water	LNCSB	0.12	0.18
Higgins 2018 (350mg ASP)	Water	LNCSB	0.04	0.18
Higgins 2019 (Asp)	SSB	LNCSB	-0.05	0.31
Higgins 2019 (RebA)	SSB	LNCSB	0	0.31
Higgins 2019 (Sacrn)	SSB	LNCSB	-0.08	0.31
Higgins 2019 (Sucralose)	SSB	LNCSB	-0.02	0.31
Madjd 2015	Water	LNCSB	0.10	0.10
Madjd 2017	Water	LNCSB	0.05	0.07
Peters 2016	Water	LNCSB	-0.20	0.09

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eData 12. Triglycerides (mmol/L)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Campos 2015	SSB	LNCSB	0.30	0.32
Ebbeling 2020	Water	LNCSB	0.03	0.07
Ebbeling 2020	SSB	LNCSB	-0.01	0.07
Ebbeling 2020	SSB	Water	-0.04	0.07
Engel 2018	Water	LNCSB	-0.05	0.23
Engel 2018	SSB	LNCSB	-0.61	0.27
Engel 2018	SSB	Water	-0.56	0.28
Hernandez-Cordero 2014	SSB	Water	-0.19	0.16
Higgins 2018 (1050mg ASP)	Water	LNCSB	-0.05	0.12
Higgins 2018 (350mg ASP)	Water	LNCSB	-0.03	0.12
Higgins 2019 (Asp)	SSB	LNCSB	-0.21	0.32
Higgins 2019 (RebA)	SSB	LNCSB	-0.13	0.33
Higgins 2019 (Sacrn)	SSB	LNCSB	-0.43	0.33
Higgins 2019 (Sucralose)	SSB	LNCSB	-0.12	0.33
Madjd 2015	Water	LNCSB	0.00	0.05
Madjd 2017	Water	LNCSB	0.04	0.06
Peters 2016	Water	LNCSB	-0.27	0.08

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eData 13. HDL-Cholesterol (mmol/L)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Campos 2015	SSB	LNCSB	0.00	0.15
Ebbeling 2020	Water	LNCSB	0.03	0.04
Ebbeling 2020	SSB	LNCSB	0.07	0.04
Ebbeling 2020	SSB	Water	0.05	0.04
Engel 2018	Water	LNCSB	-0.07	0.11
Engel 2018	SSB	LNCSB	-0.06	0.11
Engel 2018	SSB	Water	0.01	0.11
Hernandez-Cordero 2014	SSB	Water	0.05	0.06
Higgins 2018 (1050mg ASP)	Water	LNCSB	-0.03	0.11
Higgins 2018 (350mg ASP)	Water	LNCSB	-0.01	0.11
Higgins 2019 (Asp)	SSB	LNCSB	0	0.16
Higgins 2019 (RebA)	SSB	LNCSB	0.05	0.16
Higgins 2019 (Sacrn)	SSB	LNCSB	0.06	0.16
Higgins 2019 (Sucralose)	SSB	LNCSB	-0.06	0.16
Madjd 2015	Water	LNCSB	0.00	0.05
Madjd 2017	Water	LNCSB	0.00	0.04
Peters 2016	Water	LNCSB	0.06	0.05

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eData 14. Total-Cholesterol (mmol/L)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Campos 2015	SSB	LNCSB	0	0.26
Engel 2018	Water	LNCSB	-0.52	0.29
Engel 2018	SSB	LNCSB	-0.91	0.34
Engel 2018	SSB	Water	-0.39	0.33
Hernandez-Cordero 2014	SSB	Water	0.13	0.32
Higgins 2018 (1050mg ASP)	Water	LNCSB	0.1	0.21
Higgins 2018 (350mg ASP)	Water	LNCSB	0.03	0.21
Higgins 2019 (Asp)	SSB	LNCSB	-0.05	0.36
Higgins 2019 (RebA)	SSB	LNCSB	0.05	0.36
Higgins 2019 (Sacrn)	SSB	LNCSB	-0.02	0.36
Higgins 2019 (Sucralose)	SSB	LNCSB	-0.07	0.36
Madjd 2015	Water	LNCSB	0.10	0.11
Madjd 2017	Water	LNCSB	0.05	0.09
Peters 2016	Water	LNCSB	-0.16	0.09

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eData 15. Systolic Blood Pressure (mmHg)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Campos 2015	SSB	LNCSB	-3.50	5.22
Ebbeling 2020	Water	LNCSB	-2.4	1.77
Ebbeling 2020	SSB	LNCSB	-2.4	1.84
Ebbeling 2020	SSB	Water	0	1.77
Engel 2018	Water	LNCSB	-4.6	4.25
Engel 2018	SSB	LNCSB	-7.2	4.33
Engel 2018	SSB	Water	-2.60	3.85
Hernandez-Cordero 2014	SSB	Water	1.9	2.08
Peters 2016	Water	LNCSB	-4.23	1.64
Tate 2012	Water	LNCSB	0.80	2.08

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eData 16. Diastolic Blood Pressure (mmHg)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Campos 2015	SSB	LNCSB	-3.20	3.49
Ebbeling 2020	Water	LNCSB	0.8	1.49
Ebbeling 2020	SSB	LNCSB	-1.5	1.56
Ebbeling 2020	SSB	Water	-2.3	1.49
Engel 2018	Water	LNCSB	-4.4	3.21
Engel 2018	SSB	LNCSB	-6.9	3.33
Engel 2018	SSB	Water	-2.50	3.34
Hernandez-Cordero 2014	SSB	Water	0.5	1.86
Tate 2012	Water	LNCSB	0.30	1.43

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eData 17. Intrahepatocellular Lipid/Liver fat (SMD)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Campos 2015	SSB	LNCSB	-0.38	0.19
Maersk 2012	Water	LNCSB	-0.08	0.20
Maersk 2012	SSB	LNCSB	-0.47	0.21
Maersk 2012	SSB	Water	-0.46	0.21

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eData 18. Alanine Aminotransferase (U/L)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Campos 2015	SSB	LNCSB	-3.50	7.1
Ebbeling 2020	Water	LNCSB	-8.80	7.07
Ebbeling 2020	SSB	LNCSB	-13.90	7.44
Ebbeling 2020	SSB	Water	-5.10	7.57
Higgins 2018 (1050mg ASP)	Water	LNCSB	2.5	2.59
Higgins 2018 (350mg ASP)	Water	LNCSB	-0.4	2.55

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eData 19. Aspartate Aminotransferase (U/L)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Campos 2015	SSB	LNCSB	-1.50	3.25
Higgins 2018 (1050mg ASP)	Water	LNCSB	-0.30	3.06
Higgins 2018 (350mg ASP)	Water	LNCSB	0.70	3.06

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eData S20: Uric Acid (mmol/L)

Author	Treatment 1	Treatment 2	Mean Difference	Standard Error
Bruun 2015	Water	LNCSB	0.01	0.03
Bruun 2015	SSB	LNCSB	-0.04	0.02
Bruun 2015	SSB	Water	-0.05	0.02
Campos 2015	SSB	LNCSB	0.00	0.03
Ebbeling 2020	Water	LNCSB	-0.01	0.01
Ebbeling 2020	SSB	LNCSB	0.00	0.01
Ebbeling 2020	SSB	Water	0.01	0.01

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